



The evolutionary history and spectral tuning of vertebrate visual opsins

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ARTICLE INFO

Keywords:

Opsins
Spectral tuning
Evolution
Retina
Cones
Rods
Photoreceptors
Natural variation
Natural selection
Sexual selection
Adaptation
Color vision
Gene loss
Gene duplications
Diversity

ABSTRACT

Many animals depend on the sense of vision for survival. In eumetazoans, vision requires specialized, light-sensitive cells called photoreceptors. Light reaches the photoreceptors and triggers the excitation of light-detecting proteins called opsins. Here, we describe the story of visual opsin evolution from the ancestral bilaterian to the extant vertebrate lineages. We explain the mechanisms determining color vision of extant vertebrates, focusing on opsin gene losses, duplications, and the expression regulation of vertebrate opsins. We describe the sequence variation both within and between species that has tweaked the sensitivities of opsin proteins towards different wavelengths of light. We provide an extensive resource of wavelength sensitivities and mutations that have diverged light sensitivity in many vertebrate species and predict how these mutations were accumulated in each lineage based on parsimony. We suggest possible natural and sexual selection mechanisms underlying these spectral differences. Understanding how molecular changes allow for functional adaptation of animals to different environments is a major goal in the field, and therefore identifying mutations affecting vision and their relationship to photic selection pressures is imperative. The goal of this review is to provide a comprehensive overview of our current understanding of opsin evolution in vertebrates.

1. Introduction

Color vision is an extremely diverse trait amongst animals. A key driver of this diversity is the divergence of light-sensitive proteins called opsins, which are expressed in the photoreceptor cells of the retina. There are multiple subtypes of opsins, some of which are not expressed in the eye and provide non-visual functions (see (Feuda et al., 2022) for a review). However, in the context of vision, the losses and gains of opsin genes, coupled with the accumulated natural variation within opsin genes over time has given rise to vertebrate species with significant differences in visual perception. Nevertheless, the presence of different opsin subtypes in each animal does not always necessarily equate to changes in color vision. We must consider the combination and patterns of opsin subtypes together with the specific photic environment of the animal as drivers of evolution. Here we provide a resource of peak absorbances for opsin proteins, derived from spectrophotometry or predicted based on opsin sequences (Supplementary File 1A and 1B). We describe the early evolution as well as the modern diversity of opsins, emphasizing the molecular changes in opsin genes and other mechanisms that affect how opsins detect different light wavelengths. We also highlight instances of gain and loss of opsin genes throughout vertebrates and

link this diversity to natural and sexual selection pressures of different species, in addition to mechanisms of expression regulation in vertebrates.

2. The evolutionary history of vertebrate visual opsins

In extant vertebrates, proteins known as c-opsins are expressed and localized to the outer segments of ciliary photoreceptors within the retina of the eye (Fig. 1A). In order to detect light, c-opsins utilize the G_t-type phototransduction cascade, whereby the c-opsin protein is bound covalently to a vitamin A-derived chromophore, commonly 11-*cis*-retinal. The chromophore is attached to the c-opsin protein at a universally conserved K296 in the seventh transmembrane helix, via a Schiff base linkage (Palczewski et al., 2000). Upon exposure to light, the chromophore undergoes a photoisomerization to all-*trans* retinal, which drives the activation of the photopigment and the downstream phototransduction cascade (Okada et al., 2001). Starting from the first early metazoan opsins, we outline the major steps that generated modern c-opsin proteins capable of detecting different light wavelengths.

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<https://doi.org/10.1016/j.ydbio.2022.10.014>

Received 3 November 2021; Received in revised form 27 October 2022; Accepted 31 October 2022

Available online 9 November 2022

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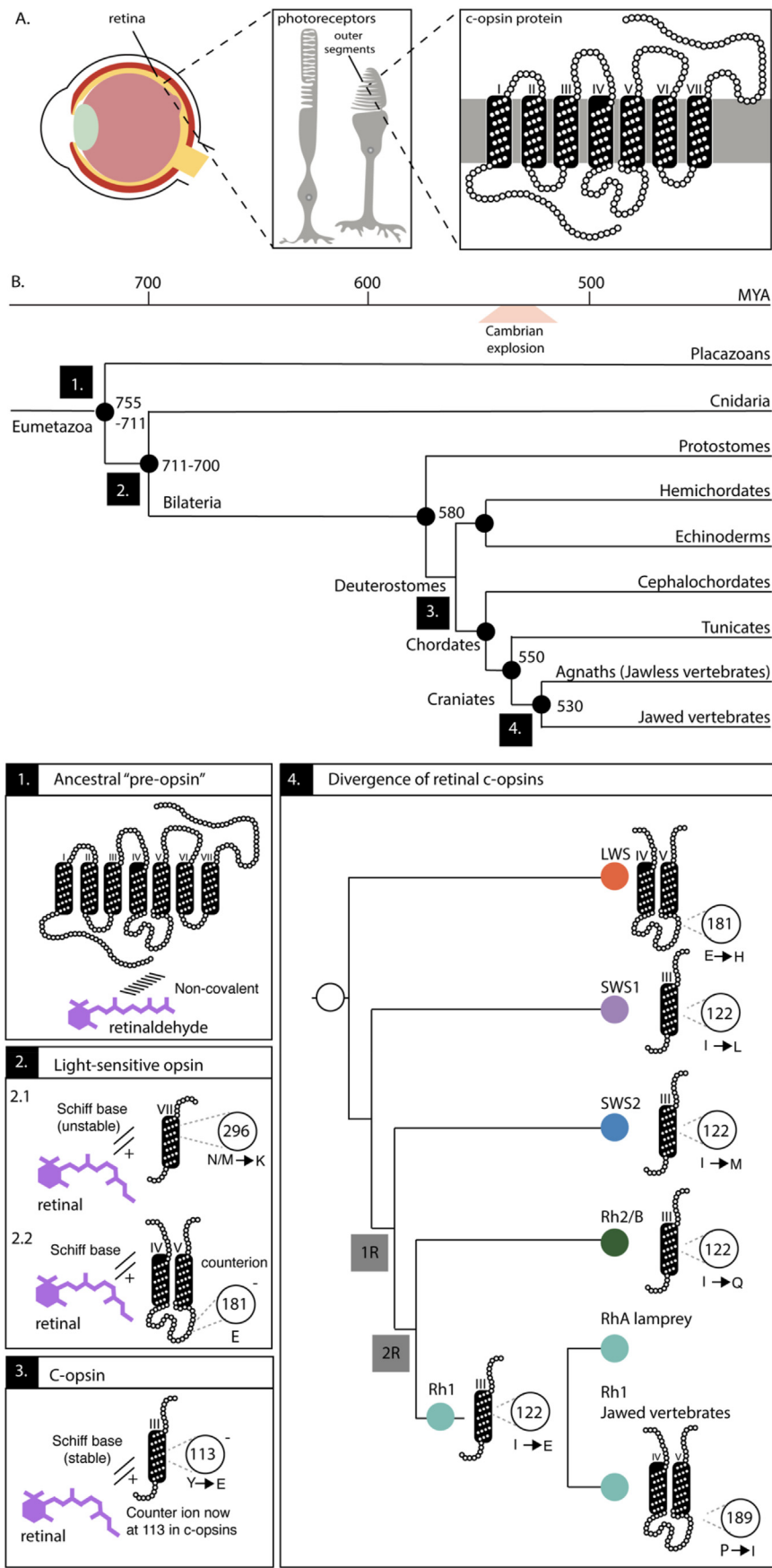


Fig. 1. The localization of extant vertebrate opsins and their early evolution. (A) c-opsin proteins are localized to the outer segments of photoreceptor cells in the retina. Transmembrane domains of the c-opsin protein are labelled (I-VII). Numbers annotated along the horizontal axis of the phylogeny indicate number of millions of years ago (MYA). (B) Steps 1 through 4 indicate significant stages in light-sensitive c-opsin protein evolution in the Eumetazoan phylogeny. **Step 1.** The “pre-opsin” GPCR bound non-covalently with the chromophore retinaldehyde. **Step 2.1.** The substitution of Asparagine (N) or Methionine (M) for lysine (K) at site 296 allowed the GPCR to bind retinaldehyde covalently via a protonated Schiff base. **Step 2.2.** Introduction of glutamic acid (E) at site 181 serves as a counterion to the protonated Schiff base (Shichida and Matsuyama, 2009). This opsin is considered light-sensitive. **Step 3.** The substitution of Tyrosine (Y) for E (Terakita et al., 2004) in the ancestral c-opsin displaced and stabilized the protonated Schiff base due to an attraction between the negatively charged E113 and positively charged K296, maintaining the opsin-chromophore dimer in a resting state (Shichida and Matsuyama, 2009; Terakita et al., 2004; Tsutsui et al., 2008). **Step 4.** The generation of multiple opsin subtypes from an ancestral opsin (white circle). The hypothesized role of 1R and 2R duplication events are indicated by grey boxes (Lamb and Hunt, 2017). Examples of variation between opsin subtypes are indicated (LWS = long wavelength sensitive (red circle)), SWS = short-wavelength sensitive opsins (SWS1 (purple circle) and SWS2 (blue circle)) and Rh = rhodopsin (Rh1/A and Rh2/B)).

2.1. Early metazoan opsins

The ancestral opsin is derived from an ancient family of proteins, the G-protein-coupled-receptors (GPCRs) (de Mendoza et al., 2014; Krishnan et al., 2012). Reconstruction of the opsin phylogeny, including non-visual opsins, predicts at least nine opsin paralogs in the ancestor to bilateria and cnidaria (Ramirez et al., 2016a). Before the divergence of placozoans and cnidarians from the bilaterian lineage (755–711 million years ago (MYA) (reviewed in (Erwin et al., 2011))), a “pre-visual opsin” GPCR bound non-covalently with the chromophore retinaldehyde (Fig. 1B, Step 1). Divergence at multiple loci gradually led to the generation of “visual opsins” that are sensitive to light, and the use of retinal as a chromophore. One critical mutation involved the introduction of Lysine (K) in place of either Asparagine (N) or Methionine (M) at site 296 in the seventh transmembrane helix (note, site number will be consistently given based on position in the bovine rhodopsin sequence (Nathans and Hogness, 1983)). This critical event occurred sometime within a 11 MY window between the divergence of the placozoans and cnidarians (Feuda et al., 2012) (711–700 MYA (Erwin et al., 2011)). This lysine residue allowed the GPCR to covalently bind retinal via a protonated Schiff base (Feuda et al., 2012) (Fig. 1B, Step 2). However, this protonated bond was highly unstable, until a second mutation arose that introduced a negatively charged Glutamic acid (E) at site 181 that serves as a counterion to the protonated Schiff base (Shichida and Matsuyama, 2009). The resultant protein is considered the ancestral, light-sensitive animal opsin (Fig. 1B, Step 2) (Feuda et al., 2012).

It has been proposed that this opsin was expressed in a primordial photoreceptor that shared morphological features common to rhabdomic and ciliary photoreceptors of extant animal species (as discussed in (Lamb, 2009; Lamb et al., 2007)). Since vertebrate vision depends on c-opsins of ciliary photoreceptors, rhabdomic photoreceptors are not the focus of this review. However, for the purposes of understanding the origin of ciliary photoreceptors and c-opsins, we will briefly describe rhabdomic photoreceptors. Rhabdomic photoreceptors are commonly found in invertebrates, and predominantly express “r-opsins” within their microvilli (Eakin, 1965; Lamb, 2013). One exception is the r-opsin melanopsin (OPN4), which is expressed in the intrinsically photosensitive retinal ganglion cells (ipRGCs) in vertebrates, and regulates circadian rhythms as well as other non-image-forming responses to light (Porter et al., 2012). Intriguingly, the ancient ancestral photoreceptor is thought to have contained both a microvilli and ciliary membrane. However, since the early opsin is thought to have been localized to the ciliary membrane of the ancestral photoreceptor, it is described as being “c-opsin-like”.

Divergence of the ancestral photoreceptor into distinct ciliary and rhabdomic subtypes containing either ciliary or microvilli membranes likely occurred in the ancestor to bilaterians, before the split of deuterostomes and protostomes, as both photoreceptor subtypes can be detected in some species from each lineage (Feuda et al., 2014; Ramirez et al., 2016b; Suga et al., 2008; Terakita, 2005; Vopalensky and Kozmik, 2009). Following the split of the deuterostome and protostome lineages, most deuterostomes lost the use of rhabdomic photoreceptors and primarily have ciliary photoreceptors within their retina. However, one theory postulates that rhabdomic photoreceptors of early deuterostomes lost their microvilli membrane structures and gained the ability to generate synapses with ciliary photoreceptors, to gradually become retinal ganglion cells (Lamb et al., 2007). This is based on the observation that both cell types express a rhabdomic class of opsin (i.e. melanopsin in intrinsically photosensitive retinal ganglion cells), share close homology in their response to neurogenetic transcription factors, and the GQ-type signaling cascade utilized by r-opsins (Arendt, 2003; Arendt et al., 2004; Lamb et al., 2007). For an extensive review of invertebrate r-opsins, see our sister review in this issue (Roberts et al., 2022).

2.2. Early chordate opsins

The rise of early chordates coincides with the onset of the Cambrian explosion (~530–540 Mya). Within this period of erupting diversity, ‘eyespot’ structures developed into primordial eyes. Such structures have been detected in extinct Cambrian species, such as trilobites (Parker, 2011). Based on fossil evidence, it is thought that these primordial eyes likely contained many of the apparatus associated with modern visual systems, including the retina, bipolar cells, lens, cornea, iris, ocular muscles, and specialized regions of brain tissue capable of processing visual information. Signaling of photoreceptors towards the brain likely became synaptic rather than chemical (though ciliary receptors are still capable of melatonin release in extant species).

Within the ancestral chordate c-opsin protein, multiple changes resulted in the expedition of light responsiveness. One important change was the displacement of the counterion from site 181 to site 113, likely caused by the substitution of Tyrosine (Y) for E in early chordates (Fig. 1B, Step 3) (Terakita et al., 2004). This displacement stabilized the protonated Schiff base due to an attraction between the negatively charged E113 and positively charged K296, maintaining the opsin-chromophore dimer in a resting state (Shichida and Matsuyama, 2009; Terakita et al., 2004; Tsutsui et al., 2008). The shift of the counterion site to E113 is thought to have reduced selection pressure on E181, since amino acid changes at site 181 are a common occurrence in many opsins of extant animals, resulting in spectral tuning of opsin subtypes towards different light wavelengths (Terakita et al., 2004).

2.3. Early vertebrates

Phylogenetic analysis suggests that the earliest c-opsins were likely more sensitive to short light wavelengths. Single-gene tandem duplication and variation that accumulated first caused the divergence of a long-wavelength sensitive opsin (“LWS”) (Lagman et al., 2013; Lamb et al., 2007; Pisani et al., 2006). The long-wavelength shift of LWS arose from the replacement of E at site 181 with Histidine (H), which gave rise to a chloride ion binding site (Davies et al., 2012; Wang et al., 1993; Yamashita et al., 2013) (Fig. 1B, Step 4).

The subsequent events that generated the range of other c-opsin subtypes (SWS1, SWS2, Rh1 and Rh2) has been a matter of debate, particularly concerning the role of whole genome duplication or block duplication events (Lagman et al., 2013; Lamb and Hunt, 2017; Larhammar et al., 2009; Nordström et al., 2004; Okano et al., 1992). Note, that although the date of the two rounds (2R) genome duplication is also a matter of debate, we adopt the conventional assumption that 2R occurred before the divergence of jawed and jawless vertebrates.

Initially, Okano et al., argued that the rod c-opsin (Rhodopsin, Rh1), was generated after the cone c-opsins (LWS, SWS1 and SWS2) had already been established (Okano et al., 1992). However, based on the synteny of the opsins (as well as other genes involved in the phototransduction cascade), it was later argued that Rh1/Rh2/SWS1/SWS2 were all generated during quadruplication of the genome during 2R (Lagman et al., 2013; Larhammar et al., 2009; Nordström et al., 2004). It was thought that quadruplication of the short-wavelength sensitive c-opsin followed by the accumulation of variation gave rise to distinct opsin subtypes; for example I122L on chromosome 7 generated SWS1, and a I122M substitution on X chromosome generated SWS2 (Fig. 1B, Step 4). Multiple copies of LWS were also likewise generated, although all but one of these copies were subsequently lost (Lagman et al., 2013; Larhammar et al., 2009; Nordström et al., 2004).

However, recent analysis of evolutionary distances between subtypes suggests that the separation of SWS1 and SWS2 pre-dates the genome duplication events (Lamb and Hunt, 2017). Then, during the first round of genome duplication (1R), the SWS2 ortholog gave rise to a Rh. Duplication of this Rh during 2R subsequently gave rise to Rh1 (rod-specific) and Rh2 (cone-specific) c-opsins (Lamb and Hunt, 2017) (Fig. 1B, Step 4). Two derived mutations in Rh1/A explain most of its rod-

specific physiological properties, including the significantly slower regeneration of opsin/chromophore intermediates of the photo-transduction cascade following photoactivation (Imai et al., 1997; Kuwayama et al., 2002). This includes the substitution at site 122 of glutamine(Q)/Isoleucine(I) for E122, which is common to both the Rh1 of jawed vertebrates and the orthologous RhA in agnaths (Collin et al., 2003b; Pisani et al., 2006)) (Fig. 1B, Step 4). In addition, I189 provides rod-like properties in the jawed vertebrate Rh1 (Collin et al., 2009; Imai et al., 1997, 2005).

With this change, the "true rhodopsin" Rh1 detects low-light levels ("scotopic vision") in jawed vertebrates, is less sensitive to differences in

color, and has high sensitivity to changes in light and dark light environments, shape, and movement. Although closely related to Rh1, Rh2 is exclusively expressed in cone photoreceptors and is cone-like in biochemical properties. The cone-opsins SWS1, SWS2, LWS, and Rh2 provide "photopic vision" in exposed light environments and are responsible for color vision. Based on parsimony of the opsin subtypes present in extant lineages, the early vertebrate ancestor likely had photic and scotopic vision, with SWS1, SWS2, LWS, Rh1, and Rh2 c-opsins in their genome (Fig. 1B, Step 4, Fig. 2).

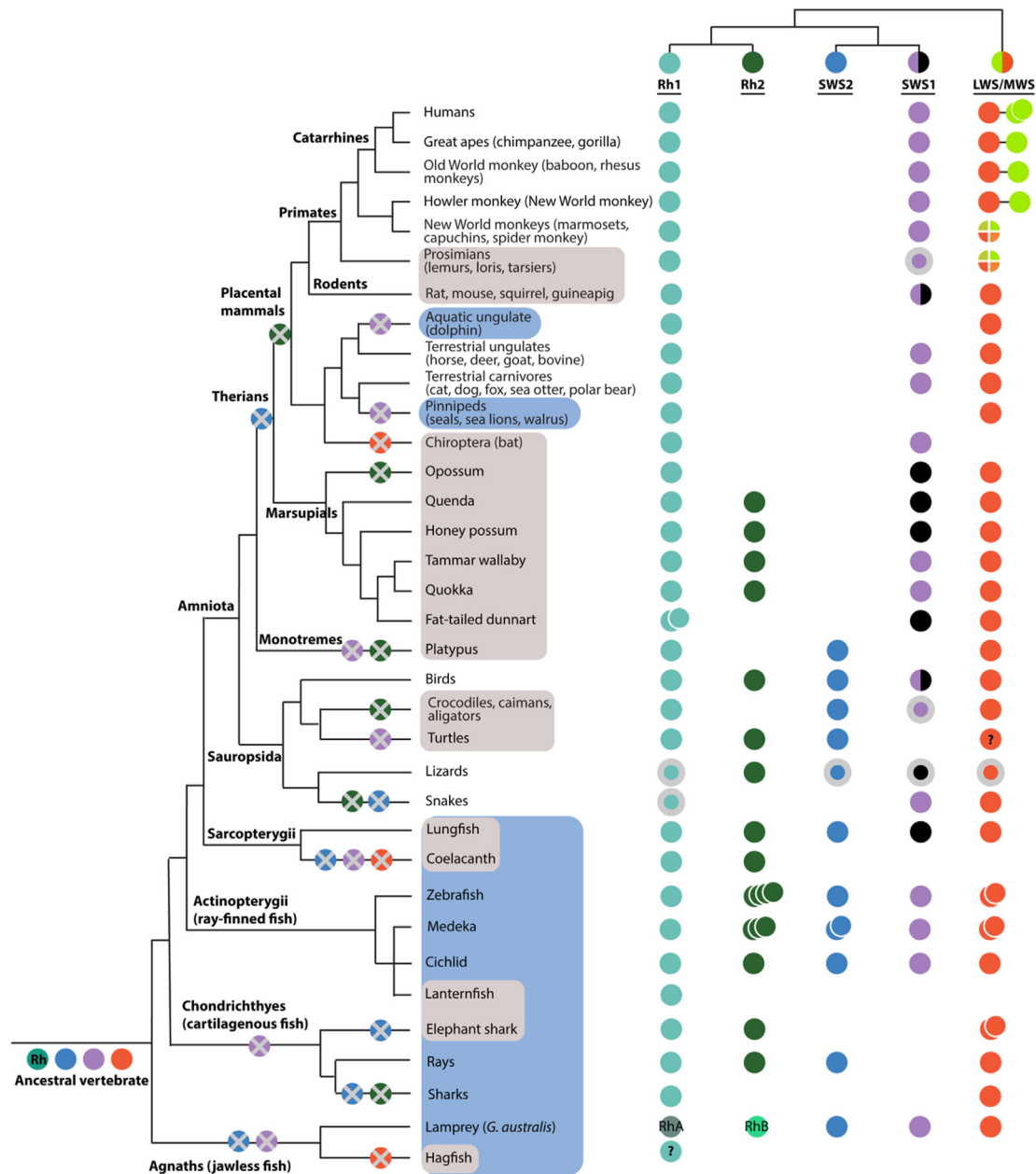


Fig. 2. An overview of c-opsin gene losses and duplications in vertebrates. Circles represent c-opsin genes, with different colors labelling different subtypes. The c-opsin gene key is as follows: Rh1(turquoise), Rh2 (dark green), SWS2 (blue), SWS1 (violet (violet-sensitive)/black (UV-light sensitive), LWS (red), MWS (light green). We illustrate the most common opsin profile, but variation of opsin loss and duplication does occur within these group. For a breakdown of the c-opsin genes within different species of each group, see [Supplementary File 1A](#). Note, the MWS found in some primates is derived from the ancestral LWS. The line between LWS and MWS in catarrhine primates and the howler monkey represent their tandem arrangement in the genome. Circles with grey outlines represent c-opsins that are lost in some species of the lineage. Single LWS/MWS opsin genes, with multiple alleles sensitive to different light wavelengths within species are represented by multi-colored circles. Species names shaded in blue or grey represent animals that live predominantly aquatic or nocturnal lifestyles, respectively. References used to compile this figure can be found in [Supplementary File 1A](#).

3. The diversity of extant vertebrate opsins

In this section, we provide an overview of opsin diversity in vertebrates. For clarity, we will now refer to c-opsin simply as "opsin". Note, that when referring to animals with mono-, di-, or tri-chromatic color vision (etc.), we refer to the number of different opsins that are expressed within the cone photoreceptors specifically (with monochromats expressing one opsin, dichromats expressing two opsins, etc.). Although vertebrate Rh1 divergence is included in this review, these are mostly expressed in rod photoreceptors which comparatively do not contribute significantly to color vision, and so are not included in "mono/di/tri/tetra-chromacy" descriptions.

Opsins of a particular subtype are maximally sensitive to distinct ranges of light. We will refer to opsin subtype alongside the light

wavelength that it is maximally sensitive to as "opsin subtype/peak absorption in nm". For example, "LWS/550" describes an opsin protein of LWS subtype, with a peak absorbance of light at 550 nm wavelength. In general, extant vertebrate opsin subtypes have peak absorption at SWS1/358–430 (UV through violet/blue light), SWS2/437–460 (violet-blue light), Rh2/470–510 (blue through green light), LWS/MWS/510–570 (green through orange/red light) and Rh1/500 (turquoise light) (Supplementary File 1A). In this section, we explain why different opsin subtypes are sensitive to different light wavelengths, and why there is a range in spectral sensitivity for each opsin subtype.

If all opsins share the Gt-type mechanism of phototransduction, how are different subtypes sensitive to (i.e. activated by) different light wavelengths? Many answers to this question lie in the sequence divergence of opsin proteins that interact with the chromophore. The amount

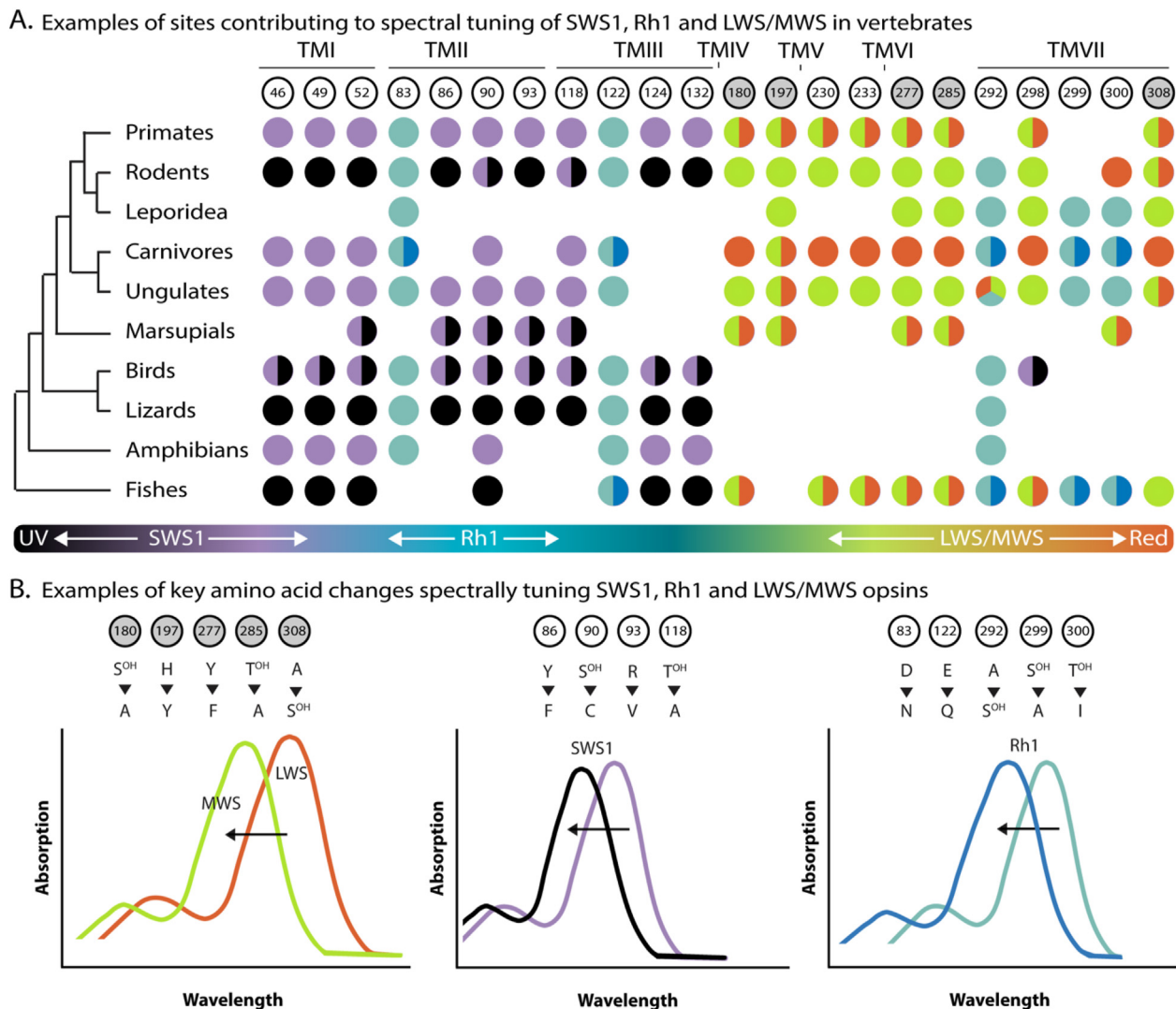


Fig. 3. Key sites that spectrally tune opsins in vertebrates. (A) The number in the top row of circles represents the position in the opsin protein-coding sequence, corresponding to the position in the bovine rhodopsin sequence (Nathans and Hogness, 1983). Each opsin is shown as a circle, with colors roughly correlating with the range of light each opsin can detect (see key at the bottom of (A), with SWS1 opsin ranging in sensitivity between UV light (shown as black circles (<400 nm)) and violet light (~400–430 nm), Rh1 = blue (~485 nm) or turquoise (~485–500 nm), MWS = green (~510–530 nm), LWS = red (~530–560 nm). Although all sites shown may contribute to tuning opsin sensitivity within each subtype, circles made up of two colors represent sites where variation led to changes in peak absorption between colors shown (e.g. variation at site 86 within birds generates SWS1 opsins sensitive to wavelengths of light from UV through to the violet spectrum). The only site that may contribute to tuning of two opsin subtypes is site 292 in ungulates, which has been shown to tune both LWS/MWS (green and red) and Rh1 (turquoise) opsin subtypes. (B) Examples of key amino acid changes along opsin protein-coding sequence in LWS/MWS (first column, red line = LWS absorbance, green line = MWS absorbance), SWS1 (middle column, black line = UV-sensitive SWS1, violet line = violet sensitive SWS1), and Rh1 (last column). Amino-acid positions shaded grey in LWS/MWS example are the sites described in the "five-site rule" (Yokoyama and Radlwimmer, 1998). Arrows show direction of amino-acid changes that are associated with decreasing wavelength required for peak absorption (with a decrease in wavelength sensitivity represented by shifting absorbance peak to the left). For a complete list of references, the peak absorption and sites that underlie opsin sensitivity in vertebrates see Supplementary File 1B.

of energy required for isomerization depends on the relative stability of either the ground state or excited state of the retinal, which is partially determined by the properties of the retinal (Sekharan and Morokuma, 2011) and the opsin protein itself (Kosower, 1988; Mathies and Stryer, 1976; Sekharan et al., 2012). Sequence divergence in the opsin protein, particularly in the amino acids near the Schiff base and the retinal binding pocket, determine the relative energy state of the molecule (Mathies and Stryer, 1976; Sekharan et al., 2012). In particular, spectral sensitivity is often affected by the presence of non-hydroxyl-bearing amino acids or hydroxyl-bearing amino acids (such as Serine (S) or Threonine (T)). For example, a hydroxyl-bearing amino acid near the Beta-ionone ring of the retinal maintains the excited state of the chromophore and requires less energy for isomerization to occur, yielding opsins that absorb long-wavelength/low-energy light (i.e., light in the red spectrum). In contrast, S/T near the Schiff-base site stabilizes the ground state of the retinal, requiring short-wavelength/high-energy light for activation (e.g., light in the blue spectrum).

The importance of hydroxyl groups in determining opsin spectral sensitivity is collectively known as the “OH-rule”, driven by the influence of OH-bearing amino acids on the interaction between the opsin protein and the retinal, specifically on electron delocalization and bond length alteration (Chang et al., 1995; Kosower, 1988; Sekharan et al., 2012). Since the substitution of OH-bearing with non-OH-bearing amino acids is highlighted as a significant driver that alters opsin protein wavelength sensitivity, we highlight OH-bearing amino acids by labelling with ^{OH}. For example, S will be referred to as S^{OH} throughout.

As all vertebrate opsins share the same mechanism of photo-transduction, it is thought that the OH-rule is conserved across species (Sekharan et al., 2012). Although not the only determining factor, mechanisms such as the OH-rule could explain why, despite a number of variable loci between opsin subtypes within and between species, only a subset of sites have been shown to contribute to differences in opsin wavelength sensitivity (Fig. 3A) (e.g. see (Yokoyama, 2008; Yokoyama and Tada, 2003)). Although many spectrally tuning sites are yet to be discovered, the illustration in Fig. 3A shows key sites commonly known to tweak opsin sensitivities of some vertebrate opsin subtypes. For a tabular view of peak absorbance and annotated loci contributing to opsin subtype tuning, see [Supplementary File 1A and 1B](#).

3.1. Non-protein coding mechanisms underlying opsin sensitivity in vertebrates

In addition to spectral tuning of opsins through molecular evolution of the protein-coding region, many animals have alternative mechanisms that regulate opsin protein activity. These include colored oil droplets, photopigments and tapetum (Fig. 4A), the differential use of chromophores (Fig. 4B), and changes in the composition of organelles in the photoreceptors. While the focus of this review is to describe opsin diversity, these mechanisms are important to highlight as they significantly affect the functionality of opsin proteins within different environments. Considering these additional factors in tandem with opsin evolution allows a better understanding of color vision across animal species.

Oil droplets: Oil droplets are spherical organelles commonly found in the inner segments of photoreceptors of many vertebrates such as diurnal lizards, turtles, and birds (Fig. 4A, bottom diagram). They improve visual acuity by enhancing the delivery of light to the cone outer segments (colorless droplets), absorbing scattered light (pigmented droplets) (Walls and Judd, 1933), protecting photoreceptors from UV irradiation (Kirschfeld, 1982), and promoting photon capture by filtering out light of specific wavelengths (Bowmaker, 1977; Roaf, 1929; Young and Martin, 1984).

The deposition of colored oil droplets varies greatly among sauropsids, acting to tweak the sensitivities of opsins by filtering light. For instance, in anoles, the sensitivities of LWS, MWS, SWS1, and SWS2 vary greatly both within and between species (Loew et al., 2002b). The differential use of oil droplets is thought to partially underlie these differences. Moreover, in turtle species like *Pseudemys scripta elegans*, oil droplets filter light that reach their four opsin subtypes (LWS/617, MWS/515, SWS2/458 and SWS1/372). Cones contain either red (LWS-expressing cones), pale green (LWS-expressing cones), orange-/transparent (LWS-, SWS2-, and Rh2-expressing cones), transparent/non-fluorescent (SWS1-expressing cones), or yellow colored oil droplets (MWS-expressing cones) (Goede and Kolb, 1994; Kolb and Jones, 1987; Ohtsuka, 1985a, 1985b). Similarly, in birds, red (R-type) oil droplets in single cones allow long wavelengths of greater than 560 nm to reach the LWS opsin. Red oil droplets are typical in herbivorous birds and less common in nocturnal birds. The oil droplets likely aid in foraging by increasing detection of the red pigments in fruits. In addition, differential filtering of light by yellow (Y-type) and transparent (T-type) oil droplets

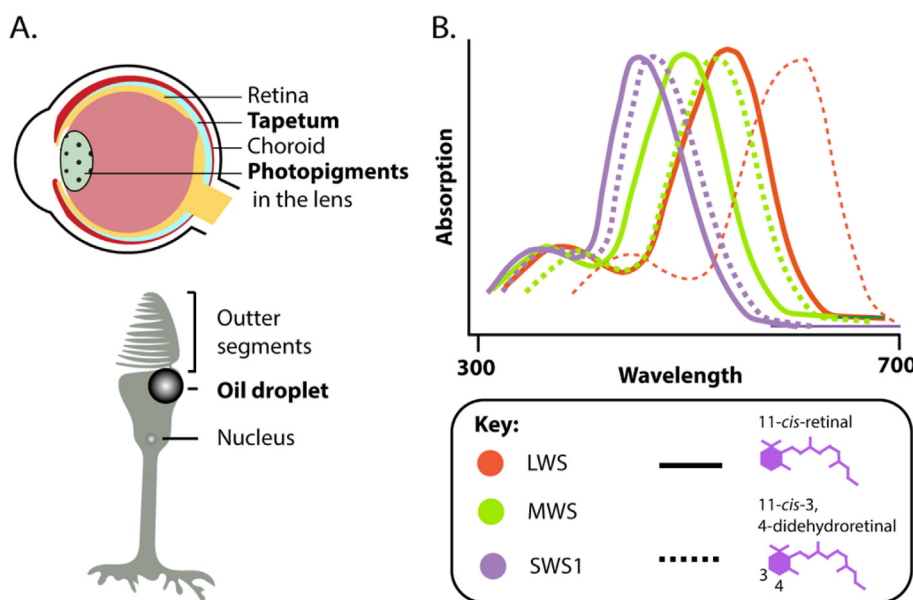


Fig. 4. Non-protein-coding mechanisms that affect opsin activity. (A) Tapetum of the choroid, photopigments within the lens (upper diagram), and oil droplets within the photoreceptors of the retina (bottom diagram) all serve to regulate the light wavelengths reaching the opsin protein (Bailes et al., 2006; Bowmaker and Knowles, 1977; Bowmaker, 2008; Bowmaker et al., 1997, 1993; Bowmaker and Martin, 1985; G. F. Cooper and Robson, 1969; G. F. Cooper and Robson, 1969; Douglas and Marshall, 1999; Hart et al., 2000a, 2000b; Hart and Hunt, 2007; Muntz, 1972; Ollivier et al., 2004; Wyman and Donovan, 1965) (Bailes et al., 2006; Douglas and Marshall, 1999; Ollivier et al., 2004; Wyman and Donovan, 1965) (B) Opsin proteins bound to different chromophores require different amounts of light energy for their activation. In general, opsins bound to 11-cis-3, 4-didehydroretinal (vitamin-A2 derived) require less energy (higher light wavelengths) than those bound to 11-cis-retinal (Vitamin-A1 derived), which require more energy (lower light wavelengths) (Alexander et al., 1994; Allen et al., 1973; Beatty, 1966; Bridges and Yoshikami, 1970; Carleton et al., 2005a; Flamarique, 2005; Hárosi, 1994; Loew et al., 2002a; Toyama et al., 2008).

occurs in bird single cones expressing SWS2, Rh2, or SWS1 opsins (Bowmaker and Knowles, 1977; Bowmaker et al., 1993, 1997; Bowmaker and Martin, 1985; Hart et al., 2000a, 2000b, 2000a). The differential distribution and spectral filtering of oil droplets between sauropsid photoreceptors has therefore contributed to the diversification of color vision within and between species.

Some animals use oil droplets to filter light exposure dynamically throughout animal lifecycles, and this can be linked to changes in foraging associated with life stages. For example, in Australian lungfish *Neoceratodus forsteri* (*N. forsteri*), yellow ellipsoid organelles within the MWS-expressing photoreceptors shift MWS peak responsiveness from 558 nm to 584 nm, as well as red oil droplets that shift LWS opsin sensitivity from 623 nm to 656 nm (Hart et al., 2008). This deposition of oil droplets exclusively occurs in adults, and modelling predicts that the upward shifts in opsin sensitivity likely assist adult *N. forsteri* to forage and distinguish between macrophyte plants.

In some cases, the use of oil droplets co-evolved with opsin spectral divergence. For example, in birds with SWS1 sensitive to longer light wavelengths, their SWS2-containing photoreceptors often contain oil droplets with a higher wavelength filtering cut-off (e.g. the oil droplet absorbs higher-wavelength light, allowing only lower wavelength light to reach the SWS2 in the outer segments) (Hart and Vorobyev, 2005). This increases the difference in light wavelengths activating the SWS1 and SWS2 opsins, which improves color discrimination between cones. Therefore, oil droplets are relevant to consider in our understanding of opsin functionality in vision. However, it is worth noting that many animals do not utilize oil droplets. Amphibians, snakes, crocodiles, and placental mammals lost the use of oil droplets due to their nocturnality (Walls, 1944).

Macular pigment and mitochondrial "microlens": Though lacking oil droplets, some placental mammals gained adaptations that are functionally convergent with oil droplets. For example, a distinctive region of the primate retina, known as the macula pigment, contains yellow carotenoids that serve to spectrally filter light reaching the retina (Snoddery et al., 1984). Moreover, tight packing of mitochondria in the cones of primates (Ball et al., 2022) and tree shrews (Knabe et al., 1997) serve as a "microlens" to focus light onto the outer segments of photoreceptors.

Similarly, the lamprey retina makes use of photoreceptor organelles to spectrally filter light reaching the opsins. The southern hemisphere lamprey, *Geotria australis*, is an andromedas species. Their metamorphosed young travel downstream towards the sea where they enter a feeding, parasitic stage by attaching themselves and feeding on fish in brightly-lit shallow waters (Hardisty and Potter, 1971a). After growing in size, they swim back upstream towards freshwater spawning grounds. During these different phases (fresh water versus saltwater), some of their photoreceptors become maximally responsive to different light wavelengths. While SWS2 is consistently sensitive at 515 nm, Rh1 and LWS peak absorptions range between 506/500 and 610/616 between downstream and upstream life cycle stages, respectively (Collin et al., 2003a). This transition in responsiveness is likely due to changes in the light wavelength reaching the opsins: in the upstream adult phase, the pigmented mitochondria-derived ellipsosome of the photoreceptors is replaced by a transparent one, which affects the spectral filtering of the cell (Collin et al., 2003a). Therefore, organelle dynamics in photoreceptors indirectly determines opsin protein activity and shapes color vision in animals.

Tapetums: Similar to oil droplets, colored retinal tapetums regulate light exposure of the opsin proteins (Fig. 4A, upper diagram). Tapetums are structural adaptation of the choroid that reflect light onto photoreceptors. Tapetums are common to many animals, particularly mammalian carnivores (Bailes et al., 2006; Douglas and Marshall, 1999; Ollivier et al., 2004; Wyman and Donovan, 1965). In some cases, differences in tapetums are thought to significantly alter the light reaching the opsin proteins. For example, canids, including dogs (*Canis familiaris*), red foxes (*Vulpes vulpes*), grey foxes (*Urocyon littoralis*), and arctic foxes (*Alopex lagopus*), are exclusively dichromatic and express LWS/∼555, and

SWS1/431–438 (as well as Rh1/501, Supplementary File 1A) (Jacobs et al., 1993; Neitz et al., 1989a). However, the colors of the tapetums vary dramatically within and between canids, ranging from light yellow-orange-green to red-yellow. Therefore, despite having opsin proteins with similar spectral sensitivities, the opsin protein activity may differ under the same light conditions due to differences in light filtering.

Lens photopigments: Similarly, colored pigments, which are common in the lenses of fishes, reptiles, marsupials, carnivores, rodents, and ungulates), regulate light exposure of opsins by filtering the light that reaches the retina (Fig. 4A, upper diagram). Deposited within the lens, yellow pigments are most common, and protect photoreceptors from short wavelength photodamage and reduce chromatic aberration due to scattering of short light wavelengths (Cooper and Robson, 1969; Jacobs, 1992; Muntz, 1972). For detailed reviews on colored lenses and tapetum see (Muntz, 1972) and (Ollivier et al., 2004).

Chromophores: Many fishes, reptiles, and anurans utilize different chromophores to tweak spectral sensitivities (Fig. 4B) (Alexander et al., 1994; Allen et al., 1973; Beatty, 1966; Bridges and Dartnall, 1972; Bridges and Yoshikami, 1970; Carleton et al., 2005b; Flamarique, 2005; Hárosi, 1994; Loew et al., 2002b; Toyama et al., 2008). While the majority of vertebrate opsins bind Vitamin A1-derived 11-*cis*-retinal, some animals use Vitamin A2-derived 11-*cis*-3, 4-Didehydroretinal, which requires a different amount of light energy to trigger the phototransduction cascade (Fig. 4B). The differential use of chromophores shifts opsin sensitivity between species and can often be linked to differences in light availability.

Crocodiles provide an excellent example. Crocodiles are expert ambushers of prey, lurking at the water's edge to target a variety of small mammals, birds, and amphibians (Lang, 1987). The crocodile visual apparatus is adapted for sensitivity in dim light environments at the water surface, including a slit pupil that regulates the amount of light illuminating the retina, which is rod-dominated intermixed with rod-like cones (Walls, 1942). Crocodiles inhabit both upstream, freshwater environments, with an abundance of longer-wavelengths of light, and saltwater environments with an abundance of shorter light wavelengths (Fig. 5A). Consistent with these differences in photic environments, the retina of the freshwater crocodile species, *Crocodylus johnstoni*, contains opsin subtypes (SWS1/426, SWS2/510, LWS/554, and Rh1/510) that are slightly shifted towards longer-wavelength sensitivity compared to the saltwater species, *Crocodylus porosus* (SWS1/424, SWS2/502, LWS/546, and Rh1/503) (Supplementary File 1A) (Nagloo et al., 2016). The longer wavelength sensitivity of the LWS/554 freshwater opsin is likely facilitated by the use of Vitamin-A2 chromophore (Nagloo et al., 2016) (Fig. 5A & B), which increases photon capture. This feature is especially useful in freshwater environments, where organic matter tends to absorb short-wavelength light and reduces the total photon exposure (Jerlov, 1976) (Fig. 5A). Therefore, the differential chromophore in crocodiles has allowed for different species to adapt to the visual needs of their environment. As a side note, the absence of UV-detecting opsins in freshwater crocodilian species is likely due to their semi-nocturnal sleep cycles and their lack of UV color patches, which are commonly used by lizards for social signaling and predator intimidation (Fleishman et al., 2011).

Interestingly, some animals switch between chromophore usage depending on light availability. For example, euryhaline teleost fish commonly experience dramatically different environments during development and migratory breeding seasons, transitioning from deep oceans to shallow, fresh-water environments and vice versa. These environmental changes pose challenges for these species, including adjustments to changes in salinity and photic environment. Many of these fish, including lampreys, eels, and salmon, cope with changes to the photic environment through chromophore switching. Flexibility in the use of vitamin A1-or vitamin A2-derived chromophores allows euryhalines to adapt when moving between environments that vary in light intensity (Alexander et al., 1994; Allen et al., 1973; Beatty, 1966; Bridges and Yoshikami, 1970; Flamarique, 2005). For example, the American eel,

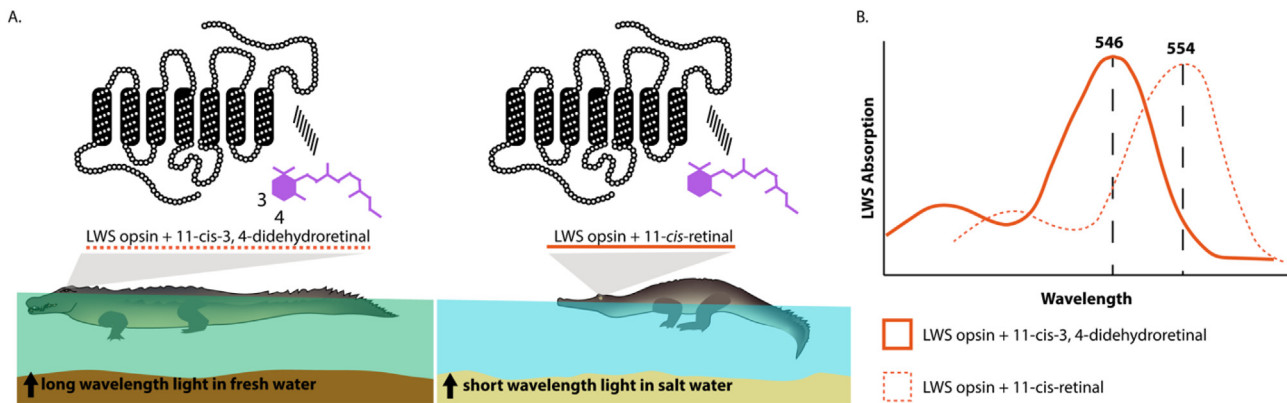


Fig. 5. The shift in LWS opsin spectral sensitivity between freshwater and saltwater crocodiles due to the use of different chromophores. (A) Freshwater environments contain an abundance of long-wavelength light, whereas saltwater environments are enriched for short-wavelength light. The LWS opsin of freshwater species binds to 11-*cis*-3, 4 didehydroretinal (Vitamin-A2 derived), whereas LWS of saltwater species binds to 11-*cis*-retinal (Vitamin-A1 derived). (B) A schematic representation of the spectral shift caused by these different chromophores, with the LWS opsin of freshwater species being sensitive to longer wavelengths than the saltwater species (554 nm versus 546 nm, respectively (Naglool et al., 2016)). (Note, this illustration is purely schematic and does not accurately represent the actual spectral shift).

Anguilla rostrata, spend their immature adult life in red/brown-tinted freshwater rivers and utilize the Vitamin A2-derived chromophore, resulting in red-shifted sensitivity at Rh1/523. They then travel to the deep salt waters of the Sargasso Sea to breed, and switch to Vitamin-A1-derived chromophore which results in blue-shifted sensitivity of Rh1 towards 501 nm (Beatty, 1975). Similarly, some cichlid species that live in clear-water habitats use A1-derived chromophores (which enables sensitivity towards shorter wavelengths), while those that dwell in turbid waters use a combination of Vitamin A1- and A2-derived chromophores (which enable shorter and longer wavelengths, respectively) (Carleton et al., 2000; Toyama et al., 2008). This flexibility tunes opsins in an environment-dependent manner and emphasizes the importance of non-opsin factors in determining color vision.

3.2. Molecular divergence of vertebrate opsins

With the non-protein-coding adaptations that affect opsin activity in mind, we now explore the diversity of opsin amino-acid sequences. We describe instances of opsin molecular evolution that have tweaked light sensitivities for some opsin subtypes, highlight cases of co-evolution, and show examples of how photic light environment shaped these trends. For example, in aquatic animals, it is more common for opsin subtypes at the extreme ends of the light spectrum (e.g. SWS1 and LWS) to have a higher degree of divergence in light sensitivities than those in the middle (e.g. Rh2 and SWS2). This is likely a consequence of their habitat, since short-wavelength (high energy) and long-wavelength (low energy) light is more likely to be absorbed or scattered by water. Throughout this section, we consider photic environment and how it might explain the divergence of opsins in extant animals.

3.2.1. Rh1

A large collection of studies of the natural selection and molecular evolution of opsins involves the adaptive radiation of Rh1. Rh1 is generally expressed in rod photoreceptors, which are mostly used in low-acuity, dim-light vision. Likely for this reason, Rh1 molecular divergence is significant in animals that experience low light habitats (e.g. deep-sea or nocturnal lifestyles). Some molecular changes appear to down-shift the spectral wavelength required for Rh1 function ("spectral tuning"), whereas others change different biochemical properties of Rh1 under low-light conditions.

Spectral tuning of Rh1: One of the clearest examples of Rh1 spectral tuning is seen when comparing fish that inhabit waters of different light environments. Presumably related to the repeated loss of cone opsins in deep-sea fish, Rh1 spectral sensitivity has diversified dramatically

between the green and blue spectrum (Fig. 6). Deep sea fishes most commonly express Rh1 with a blue-shift compared to non-deep-sea species (deep-sea species shaded in grey in Fig. 6). In many deep sea species, the blue-shift in deep-sea Rh1 sensitivity was likely naturally selected for, as the only light reaching deep sea (e.g. 320–850 m deep) is short-wavelength residual sunlight and bioluminescence. As sunlight passes down to these depths, the water column effectively filters out many wavelengths, leaving only light between 460 and 490 nm (Jerlov, 1976; Kampa, 1970). Based on these conditions, it is unsurprising to learn that the majority of deep sea fishes have lost cone cells and cone opsin genes. Instead, a wide array of rhodopsins that are tuned to shorter light wavelengths have arisen (Fig. 6).

The Rh1 blue-shift has been achieved by convergent evolution, whereby amino acid substitutions occurring at a possible three to 11 key sites in the opsin sequence generated Rh1 proteins sensitive to 531 - 468 nm light (Archer et al., 1995; Archer and Hirano, 1997; Bowmaker et al., 1988; Cornwall et al., 1989; Crescitelli, 1956; De Busserolles et al., 2015; Denton and Warren, 1957; Douglas and Partridge, 1997; Fernandez, 1979; Hisatomi et al., 1991; Hunt et al., 2001; Johnson et al., 1993; Musilova et al., 2019; Partridge et al., 1988; Schwanzara, 1967; Yokoyama, 1995, 2000a). In bony fishes, these sites include 83, 122, 124, 132, 208, 261, 292, 299 and 300 (Supplementary File 1B, Fig. 6), which involves the substitution of hydroxyl-bearing with non-hydroxyl-bearing amino acids, as well as the recently discovered sites 111 and 118 (Musilova et al., 2019).

For example, site 292 is a key site for Rh1 tuning with a high degree of natural variation between fish species (Fig. 6). Rh1 alleles with 292A is more common in fishes inhabiting shallow waters, while 292S^{OH} are more common in deep-sea species (Fig. 6). Within cichlids alone, amino-acid turnover at site 292 occurs very frequently throughout the phylogeny (Nagai et al., 2011). Species have undergone rapid adaptive radiation to inhabit deep or shallow water environments, with the A292S^{OH} substitution occurring up to four times, while S^{OH}292A has occurred up to 3 times throughout the cichlid lineage. A recent study of Rh1 divergence in Tanganyikan cichlids confirmed the correlation between amino-acid changes at sites 292 and 299, and the depth of water that the species inhabit (Ricci et al., 2022).

In contrast to these deep-sea species, the last common ancestor to the dragonfish *Aristostomias tittmani* and *Malacosteus niger* displays a red-shift in Rh1 sensitivity likely due to F261Y and S^{OH}292E substitutions (Fig. 6, Supplementary File 1). This adaptation may contribute to the detection of far-red bioluminescence which is commonly emitted by deep-sea species, along with the use a bright red tapetum that reflects long light wavelengths into the eye within these species (Bowmaker et al., 1988; Somiya,

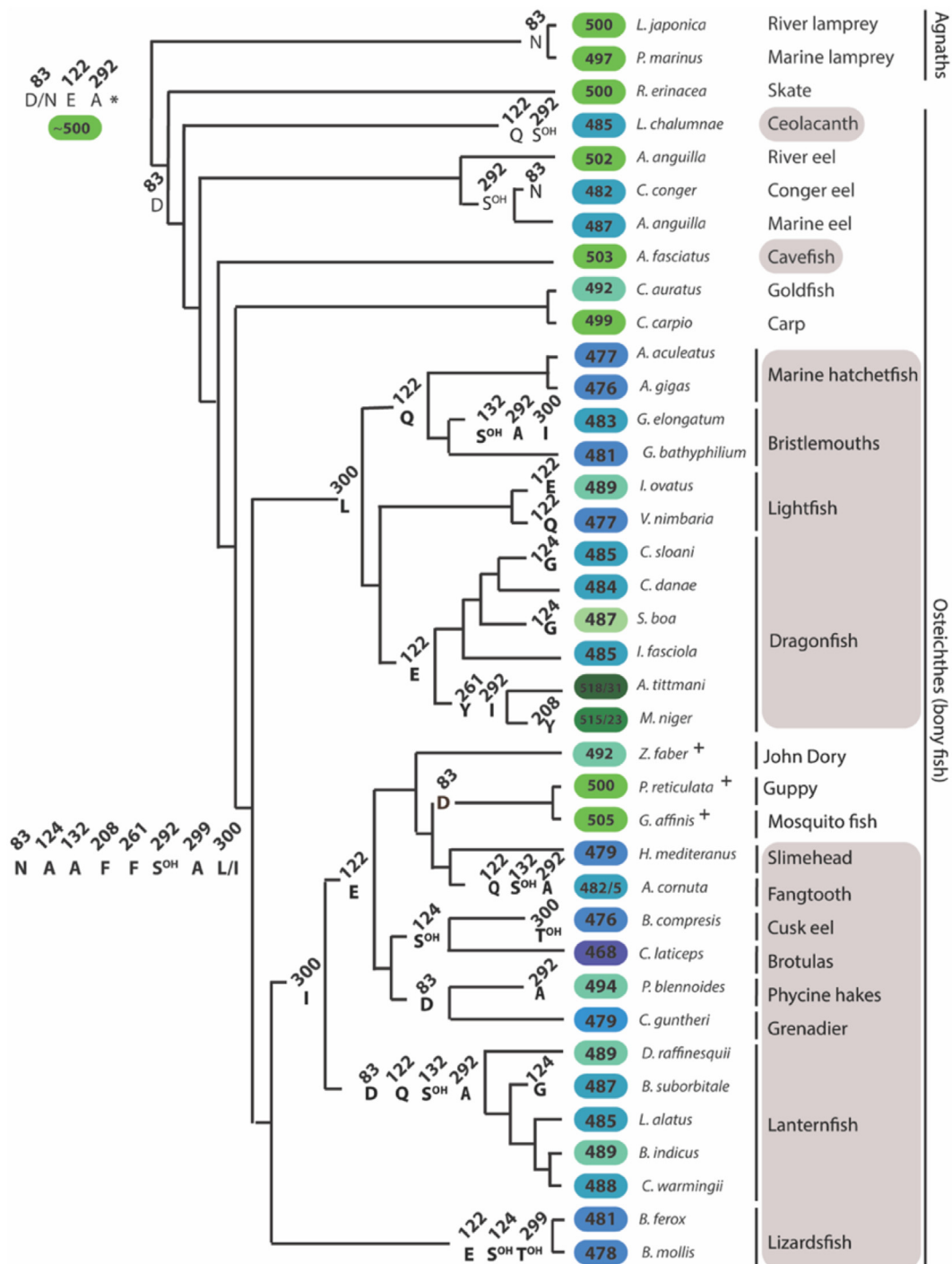


Fig. 6. Spectral tuning of Rh1 in fishes. Up to nine key sites that contribute to Rh1 spectral tuning. The amino acids in the ancestral Rh1 (indicated by *) was predicted based on parsimony, though sites 83 were difficult to interpret considering the distribution of D/N83 in extant fishes (Supplementary File 1C). Predicted retention of amino acids from the last common ancestor is depicted by a dashed line and substitutions show the newly introduced amino acid throughout the phylogeny schematic. Known or predicted wavelength sensitivities are indicated in colored spheres on the right. Different colors are used to represent down-shifted or up-shifted wavelengths and do not reflect the exact change in wavelength sensitivity. Amino acids and peak absorption wavelengths were taken from (Bowmaker et al., 1988; De Busserolles et al., 2015; Denton and Warren, 1957; Douglas and Partridge, 1997; Fernandez, 1979; Hunt et al., 2001; Partridge et al., 1988) (Supplementary File 1B). Fishes living predominantly in low-light environments are shaded in grey.

1982). Therefore, natural selection has driven diversification of Rh1 within fishes towards different wavelength sensitivities.

In contrast to fish Rh1, relatively little spectral variation is detected in mammalian Rh1 proteins (Fig. 7). The majority of mammals express Rh1 sensitive to ~500 nm light, likely determined by the high degree of conservation of amino acids at four key sites: D83, E122, A292 and A/

S^{OH}299 (Supplementary File 1B, Fig. 7). Nevertheless, variation is observed most commonly in aquatic species, such as the northern elephant seal, which inhabits deep-sea environments and has down-shifted Rh1 sensitivity, likely caused by two substitutions at the four key sites (Fig. 7). In the last common ancestor to the northern elephant seal, leopard seal, weddell seal, and harp seal, a derived S^{OH}299A

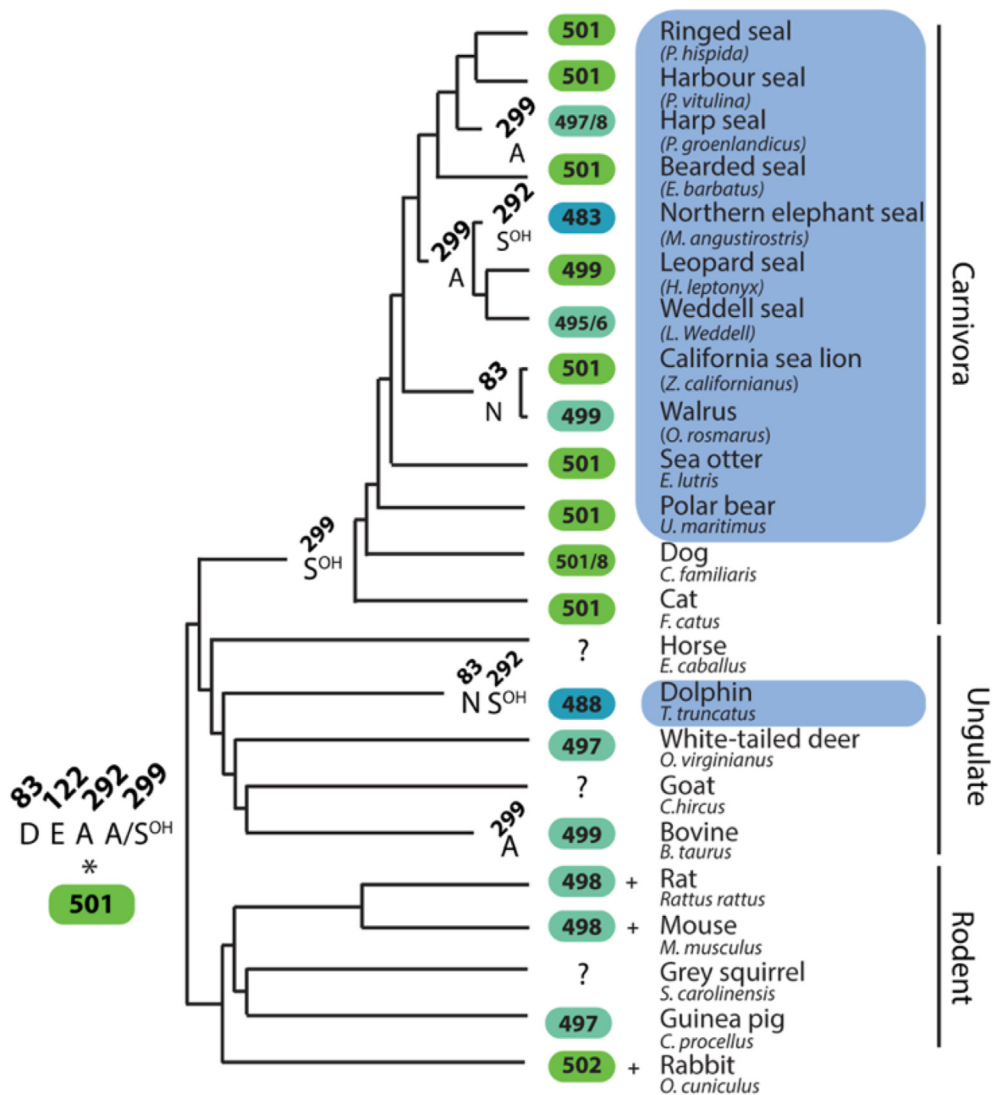


Fig. 7. Spectral tuning of Rh1 in non-primate mammals. Four key sites are thought to spectrally tune Rh1 (83, 122, 292, 299) in the non-primate placental mammals shown. The ancestral opsin (*) was predicted based on parsimony (Supplementary File 1C), though some sites such as 299 of Rh1 were difficult to interpret considering the degree of variation in extant mammals at these sites. Retention of amino acids from the last common ancestor is depicted by a dash. Substitutions show the newly introduced amino acid. Question marks represent species where opsin peak absorbance is unknown. Blue indicates down-shifted, and green indicated up-shifted wavelengths in Rh1. + symbols indicate species where the not all amino acids are known at these key sites. Colors do not truly reflect the exact change in spectral sensitivity. Species names shaded in blue range spend significant amounts of time in aquatic environments. Note, Rh1 peak absorbance has not been analyzed for Rh1 in horse, goat, pig, sheep, grey squirrel, or guinea pig. Amino acid sequences and known or predicted peak absorption wavelengths for Rh1 in each species can be found in Supplementary File 1A and 1B and were taken from (Bridges, 1959; Crescitelli, 1958; Fasick et al., 1998; Fasick and Robinson, 2000; Hunt et al., 2001; Jacobs, 1993; Lavigne and Ronald, 1975; Levenson et al., 2006a; Lythgoe and Dartnall, 1970; Newman and Robinson, 2005; Southall et al., 2002; Yokoyama, 2000b).

substitution occurred (Levenson et al., 2006a; Lythgoe and Dartnall, 1970; Southall et al., 2002) (Fig. 7). In combination with a substitution of A292S^{OH} in the northern elephant seal Rh1, spectral sensitivity is blue-shifted to 483 nm. The down-shift in the elephant seal Rh1 compared to other seals could be linked to their tendency to dive deeper than other pinnipeds. The same substitution of A292S^{OH} also occurred independently in dolphins (alongside a N83 substitution (Fasick et al., 1998; Yokoyama, 2000b)), in addition to many deep-sea fishes (Fig. 7). The reoccurrence of A292S^{OH} in these different lineages provides an example of convergent evolution of Rh1 in aquatic animals across vertebrates, driven by light availability.

In addition to site 292, site 83 deserves special attention. As shown in the examples in Fig. 7, substitution of D83N in Rh1 is associated with decreasing the sensitivities, and is most often detected in animals inhabiting lower-light environments (e.g. in aquatic mammals such as dolphins (Fasick et al., 1998) (Fig. 7), whales (Dungan and Chang, 2017), deep-sea fish species (Fig. 6, see Supplementary File 1B for citations) and eutherian and placental mammals like echidna and bats (Bickelmann et al., 2012)). D83N is also associated with improving low-light adaptation by affecting downstream mechanisms in phototransduction. For example, 83N accelerates the formation of the Rh1 active signaling state (Sugawara et al., 2009), while increasing the rate at which the retinal chromophore is released following the isomerization event (Bickelmann et al., 2012). This increases the length of time that Rh1 is in an active

state, increasing Rh1 light sensitivity. Therefore, molecular variation in opsins can alter properties of opsin function other than the wavelength of peak absorbance.

Molecular adaptation of Rh1 driven by stressful environments: In addition to the molecular divergence of Rh1 that tunes spectral sensitivity, some changes arose for non-visual reasons. For instance, L59Q and M288L substitutions in the Andean and Amazonian catfishes, which live at very high altitudes, are adaptive as they allow for accelerated Rh1 protein kinetics (Castiglione et al., 2017). Similarly, amino acid changes at sites 156, 196, 213, and 275 in deep sea fishes increase the opsin protein stability under high hydrostatic pressure by reducing the compressibility of the opsin dimer (Porter et al., 2016). Therefore, in addition to photic selection pressures, opsins also adapt to other environmental stressors to ensure vision when animals inhabit challenging environments.

Molecular evolution of Rh1 to provide cone-like properties: Intriguingly, variation within some vertebrate Rh1 proteins has led to more cone-opsin-like properties. For example, in the cone-exclusive retina of the chameleon (*Anolis carolinensis*), Rh1 is expressed and functions within cone cells (Kawamura and Yokoyama, 1994). *A. carolinensis* express the opsins (SWS1/358–365, SWS2/437–462, Rh2/491–503, LWS/560–625) as well as rhodopsin (Rh1/491–496) in cones (Supplementary File 1A) (Kawamura and Yokoyama, 1998; Loew et al., 2002a). The adaptation of Rh1 is proposed to have occurred through mutation of at least one of the 6 divergent amino acid sites at positions 22, 155, 159, 199, 232, and 319

(Supplementary File 1B)(Kawamura and Yokoyama, 1994; Yokoyama, 2000b).

Conversely, in anurans (frogs and toads), the traditionally cone-exclusive opsin SWS2 is expressed in rod photoreceptors ("green rods") (Hisatomi et al., 1999; Ma et al., 2001). Intriguingly, these green rods have also been coined "super rods", which have a significantly reduced rate of phototransduction quenching compared to the Rh1-expressing rods. For this reason, green rods are thought to be an adaptation of these generally nocturnal animals, functioning as highly specialized receptors under low-light conditions (Astakhova et al., 2021).

Interestingly, these rod-to-cone trends in evolution are not confined to the opsin proteins themselves. For example, in sauropsids that have adapted to crepuscular lifestyles (mostly active at dawn and dusk), their cones are in intermediate-stages of "transmutation", whereby the cones are transmuting into rod photoreceptors (Simões et al., 2016; Walls,

1934, 1942). Some species fully re-adapted to nocturnal environments, such as *Hypsiglena* snakes, with retinas full of transmuted rods that completely lack cone photoreceptors (Simões et al., 2016). The transmutation of photoreceptor subtypes also occurred in nocturnal lizard species, including the Tokay gecko (*Gekko gekko*) and the Schlengel Japanese gecko (*Gekko japonicus*), which express the cone opsins LWS, SWS1, and Rh2 in rod/cone intermediate cells (Kojima et al., 1992; Liu et al., 2015; Loew, 1994; Röhl, 2000; Yokoyama and Blow, 2001) (Supplementary File 1A). Despite having morphological features associated with cones, these photoreceptors exhibit rod-like electrical response profiles, which is likely explained by divergence in the phototransduction pathway (Kojima et al., 1992). As animals inhabit an incredible range of photic environments, both Rh1 and the photoreceptor are heavily influenced by photic availability. The evolution of Rh1 in response to changing light environments is convergent across vertebrates

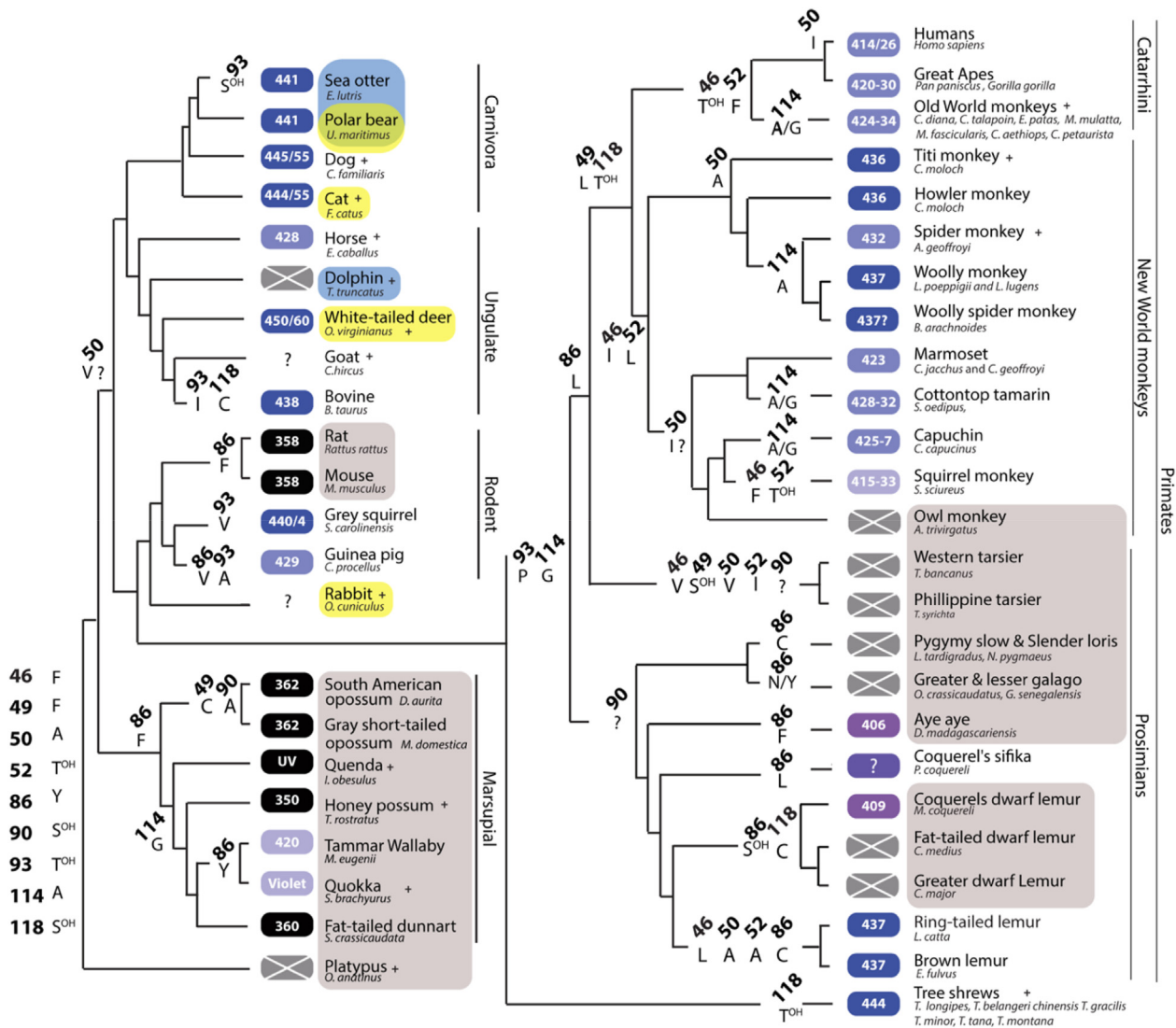


Fig. 8. Spectral tuning of SWS1 in mammals. At least nine sites spectrally tune SWS1 in mammals (46, 49, 50, 52, 86, 90, 93, 114, 118). The ancestral mammalian opsin (*) was predicted based on parsimony (Supplementary File 1C). Substitutions show the newly introduced amino acid. Peak absorbance of each opsin is shown in colored spheres in nm, with grey crossed-out spheres indicating loss of functional SWS1. Question marks, or + represent species where either the opsin sensitivity, or all of the amino acid sequence is unknown, respectively. Colors do not truly reflect the exact change in spectral sensitivity. Species names shaded in blue spend significant amounts of time in aquatic environments, grey shading indicates nocturnal species, yellow indicates crepuscular species. Amino acid sequences and known or predicted peak absorption wavelengths for SWS1 can be found in Supplementary File 1A and 1B, and were taken from (Arrese et al., 2005; Blakeslee et al., 1988; Bowmaker et al., 1991, 1980; Carvalho et al., 2006, 2012; Chiu et al., 1994; Cowing et al., 2008; Deeb et al., 2003; Hemmi and Grünert, 1999; Hiramatsu et al., 2005; Hunt et al., 2009; Ibbotson et al., 1992; Jacobs, 1993; Jacobs et al., 1996; Jacobs and Deegan, 2003, 1994; Kawamura and Kubotera, 2003; Levenson et al., 2006a; Mantovani et al., 2020; Merbs and Nathans, 1992; Mollon et al., 1984a; Shi et al., 2001; Shi and Yokoyama, 2003; Shyue et al., 1998; Strachan et al., 2004; Tan et al., 2005; Tan and Li, 1999; Travis et al., 1988; Yokoyama and Radlwimmer, 1999).

experiencing similar selection pressures, whilst under the constraints imposed by maintaining functionality.

3.2.2. SWS1

As shown in Fig. 3A, the majority of sites affecting SWS1 spectral tuning lie in the first three transmembrane domains of the protein, between sites 46 and 132. Variation at amino acids in these domains spectrally tune SWS1 between UV (below ~ 400 nm) and violet light wavelengths in vertebrates. In the following sections, we discuss the divergence of SWS1, which like Rh1 shows trends of molecular evolution in response to changing light environments (such as those imposed by nocturnality), in addition to complex social signaling.

SWS1 spectral tuning: One of the most striking trends we have gathered from our review of vertebrate opsins is the link between SWS1 spectral tuning and nocturnality. In general, nocturnality is associated

with UV-sensitivity of SWS1 or loss of functional SWS1 (nocturnal species shaded in grey, Fig. 8). For example, marsupials, some rodents, bats, and some prosimian primates are nocturnal, with UV- or near UV-sensitive SWS1 (Fig. 8). In mammals, at least nine amino acids underlie spectral tuning of SWS1 opsin (Fig. 8). Variation at these sites generated SWS1 opsins sensitive to light from the UV to the violet/blue spectrum. Based on parsimony, we predict that the last common ancestor to mammals had SWS1 with F46, F49, A50, T^{OH}52, Y86, S^{OH}90, T^{OH}93, A114 and S^{OH}118 (Supplementary File 1C and Fig. 8). Since these amino acids are conserved in the polar bear (except site 50, in which the amino acid is unknown), we predict that the ancestral mammal SWS1 was sensitive to ~ 441 nm (Levenson et al., 2006a). Two key sites that have conserved effects throughout vertebrate SWS1 are sites 86 and 90 (Cowling et al., 2002; Shi et al., 2001; Tada et al., 2009). For example, SWS1 of teleosts, marsupials, rodents, birds, and primates with F86 and/or S^{OH}90 are

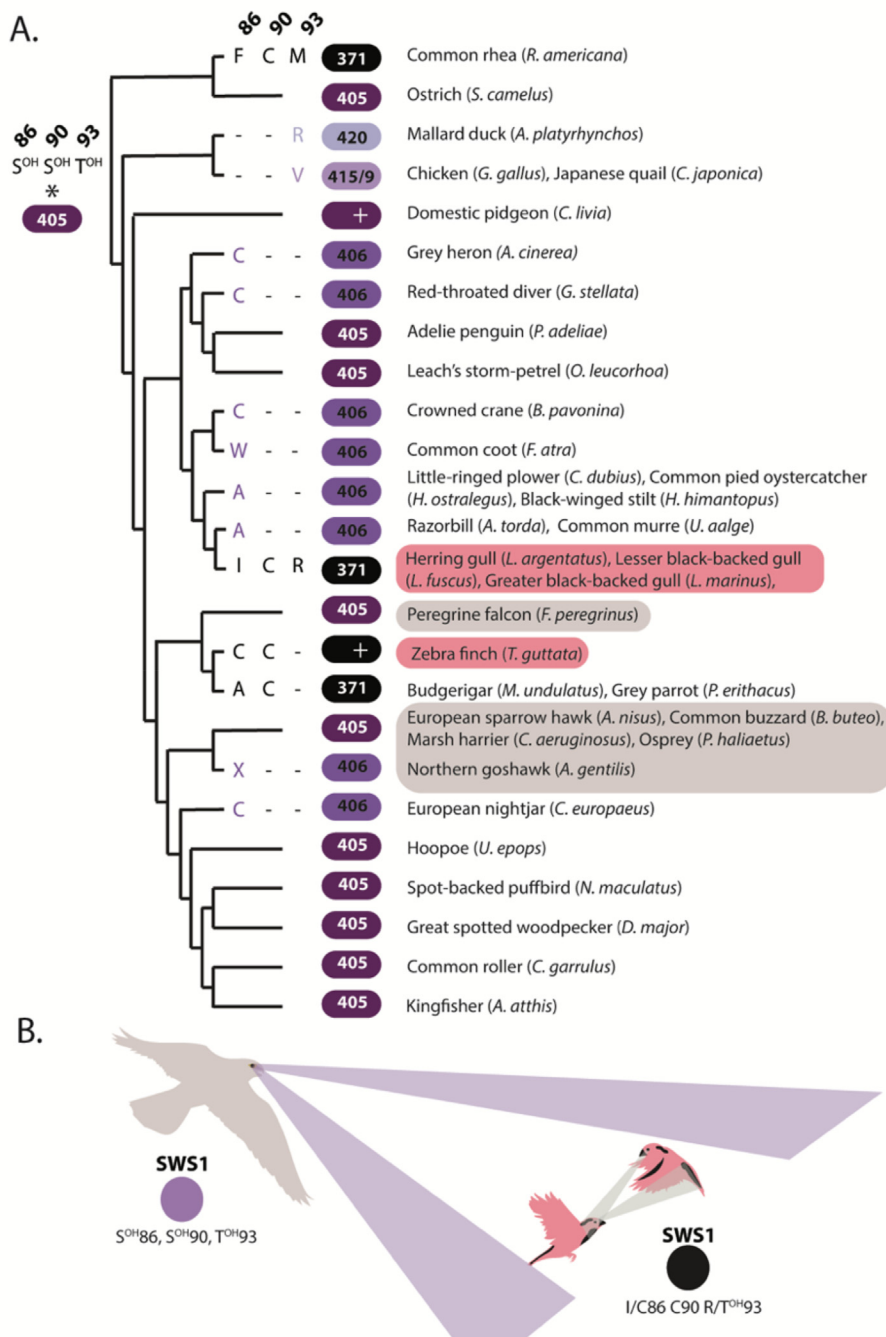


Fig. 9. Spectral tuning of SWS1 and predator-prey relationships in birds. (A) Three key sites contribute to SWS1 spectral tuning of birds (86, 90 and 93). The ancestral SWS1 sequence (*) was predicted based on parsimony, with a predicted sensitivity to ~405 nm light based on this sequence (Supplementary File 1C). Retention of amino acids from the last common ancestor is depicted by a dash. Substitutions show the newly introduced amino acid. Different colors are used to represent more UV-shifted (black) or violet wavelengths (light violet), and do not truly reflect the exact wavelengths that can be detected. (Note, the transition between UV light and visible, violet light is at ~400 nm). + indicate instances where SWS1 peak absorbance is variable in the literature, likely due to differences in peak absorbance measurements. Raptor bird species are shaded in grey, while passerines and Psittaciformes are shaded in pink. Amino acid sequences and known or predicted peak absorption wavelengths for SWS1 in each species can be found in Supplementary File 1A and 1B and were taken from (Bowmaker et al., 1997; Bowmaker and Martin, 1985; Hart et al., 1998, 2000a; Jane and Bowmaker, 1988; Maier and Bowmaker, 1993; Ödeen and Håstad, 2003; Okano et al., 1992; Wilkie et al., 2000). (B) Passerines (pink birds) often have UV-sensitive SWS1, due to I/C86, C90 and R/T^{OH}93 amino acids in the protein sequence. UV coloration in passerines (black patterns) is thought to allow for social signaling, without detection from predators raptors (grey bird) that cannot detect UV light (having violet-sensitive SWS1 with S^{OH}86, S^{OH}90, T^{OH}93) (Bowmaker, 1977; Bowmaker et al., 1997; Ödeen and Håstad, 2003; Okano et al., 1992; Shi and Yokoyama, 2003; Wilkie et al., 2000).

generally sensitive to light in the UV spectrum (e.g. (Carvalho et al., 2012; Johnson et al., 1993; Mitchell et al., 2021; Shi and Yokoyama, 2003; Yokoyama, 2000b).

A Y86F substitution likely occurred in the last common ancestor to marsupials, in the last common ancestor to mice and rats (Arrese et al., 2002, 2005; Carvalho et al., 2012; Shi and Yokoyama, 2003), in some prosimian primates (Carvalho et al., 2012), and birds (Fig. 9A). In primates, Y86F/S^{OH} appears to down-shift opsin sensitivity in aye ayes and Coquerel's dwarf lemurs towards near-UV wavelengths, while C86 (in combination with substitutions F46L, V50A, T^{OH}52A) gave rise to SWS1 opsin sensitive to long wavelengths in the ring-tailed and brown lemur (Fig. 8). Similarly, in some nocturnal bat species, SWS1 UV sensitivity is likely achieved by T52, F86, T^{OH}93, A114, and S^{OH}118 amino acid combinations (Wang et al., 2004). Therefore, variation occurring at site 86 is a key determinant of SWS1 spectral tuning in many vertebrates, and the Y86F substitution is an example of convergent evolution, arising independently in multiple nocturnal vertebrate groups to spectrally tune SWS1 between the UV and violet light spectrum (Fig. 8). The evolution of UV-sensitive SWS1 opsin has different roles depending on the species. In many marsupials, UV-sensitivity may be related to their tendency to be most active during low-light hours of the day such as twilight, when UV light wavelengths are more prevalent (Anich et al., 2021; Meisner, 1983;

Pine et al., 1985). In bats, UV-sensitive SWS1 opsin is attributed to the advantages in flower and insect detection (Müller et al., 2009; Speakman, 2001).

In birds, divergence of SWS1 opsin sensitivity plays an integral role in social signaling within and between vertebrate species (Ödeen and Håstad, 2003). Bird SWS1 sensitivities range between UV and violet light and are likely determined by variation at sites 86, 90, and 93 (Fig. 9A). Avian SWS1 opsin is thought to have switched between UV and violet light sensitivity at least 14 times over the course of bird divergence (Ödeen and Håstad, 2003). Many bird species including predatory raptors have a S^{OH}90 T^{OH}93 amino acid combination in SWS1 driving peak sensitivity to violet light (~405 nm) (shaded in grey, Fig. 9A) (Bowmaker, 1977; Bowmaker et al., 1997; Ödeen and Håstad, 2003; Okano et al., 1992; Shi and Yokoyama, 2003; Wilkie et al., 2000). In contrast, raptor prey, including passerines and some Psittaciformes, significantly down-shifted the sensitivities of their SWS1 opsins into the UV spectrum (pink shading, Fig. 9A) (Bowmaker et al., 1997; Ödeen and Håstad, 2003; Shi and Yokoyama, 2003; Wilkie et al., 2000). These animals contain SWS1 with I/C86 C90 R/T^{OH}93 (pink shading, Fig. 9A). These differences in vision between raptors and their prey likely have interesting consequences for the evolution of their behaviors. Since raptors cannot detect UV light yet their prey retain UV vision, it has been proposed that

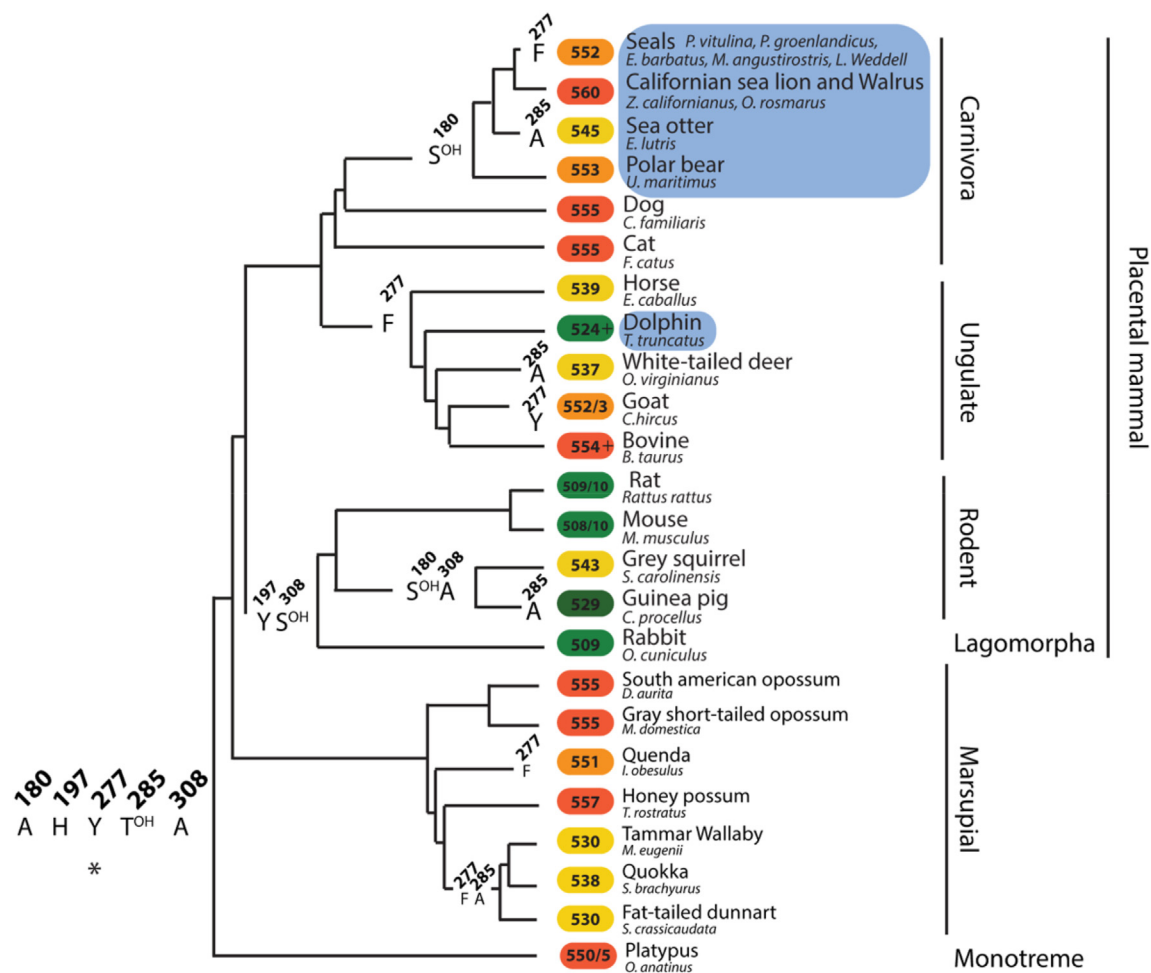


Fig. 10. Spectral tuning of LWS/MWS in non-primate placental mammals. Five key sites are through to spectrally tune LWS/MWS (180, 197, 277, 285, 308) in the non-primate placental mammals shown. The ancestral opsin (*) was predicted based on parsimony (Supplementary File 1C). Retention of amino acids from the last common ancestor is depicted by a dash. Substitutions show the newly introduced amino acid. Green and red indicates down- and up-shifted wavelengths in LWS/MWS. + symbols indicate species where the not all amino acids are known at these key sites. Species names shaded in blue range spend significant amounts of time in aquatic environments. Note, the sequence of LWS/MWS has not been analyzed for LWS/MWS of the rat, dolphin, or bovine. Amino acid sequences and known or predicted peak absorption wavelengths for LWS/MWS in each species can be found in Supplementary File 1A and 1B and were taken from (Carroll et al., 2001; Fasick et al., 1998; Jacobs et al., 1993; Levenson et al., 2006a; Neitz et al., 1989b; Newman and Robinson, 2005; Sun et al., 1997; Wienrich and Zrenner, 1984b; Yokoyama and Radlwimmer, 1999).

UV-based social signaling in passerines and some Psittaciformes has been selected for, given the reduced cost of signaling on detection by predators (Ödeen and Håstad, 2003)(Fig. 9B). Therefore, the evolution of SWS1 opsins alters the interactions between predators and prey, enabling the radiation of social signaling within and between species.

In summary, multiple different selective pressures, including light availability, UV coloration in social signaling, and diet have shaped the spectral tuning of SWS1 towards visible or UV light wavelengths. However, the convergence of SWS1 towards UV light sensitivity across many vertebrates appears to have been achieved through independent coevolution of the amino acid at the same key sites.

3.2.3. LWS/MWS

In contrast to SWS1, the key sites that spectrally tune LWS/MWS opsins are located between amino acids 180 to 308, within transmembrane domain IV, as well as cytoplasmic and intracellular domains. In many mammalian LWS/MWS opsins, the “five site rule” predicts much of the spectral sensitivity differences (i.e. amino acid substitutions occurring at sites 180, 197, 277, 285 and 308) (Yokoyama and Radlwimmer, 1998) (Supplementary File 1B). Based on parsimony, we predict that the ancestral mammalian LWS opsin contained A180, H197, Y277, T^{OH}285, and A308 (Fig. 10). This LWS was likely sensitive to ~555 nm light, as many extant mammals retained this allele (such as dog, cat, opossums, honey possum and platypus) (Arrese et al., 2002; Hunt et al., 2009; Jacobs et al., 1993; Neitz et al., 1989b; Sumner et al., 2005; Wakefield et al., 2008; Wienrich and Zrenner, 1984a; Yokoyama and Radlwimmer, 1999).

In particular, site 277 underwent repeated amino acid turnover to either increase or decrease wavelength sensitivity in mammals (Supplementary File 1A and 1B, Fig. 10). We predict that the last common ancestor of the tammar wallaby, quokka, and fat-tailed dunnart attained Y277F, which downshifted LWS to mid-range light levels (530–538 nm) (Arrese et al., 2002, 2005; Cowing et al., 2008; Deeb et al., 2003; Strachan et al., 2004) (Supplementary File 1A and 1B, Fig. 10). The down-shifting Y277F substitution also occurred independently in the quenda (LWS/551), in the last common ancestor of seals (LWS/552), at the base of the ungulate lineage (giving rise to an allele (A180, H197, F277, T^{OH}285, A308) that was retained in horse, MWS/539) (Carroll et al., 2001; Yokoyama and Radlwimmer, 1999), in some bat species such as the microbat *M. velifer* (LWS/502) (Wang et al., 2004), and in many rodents with down-shifted MWS/LWS (such as the Degu, Alpine marmot, Golden hamster, Jerboa and Mongolian gerbil, and the Elephant shrew (502, 516, 504, 490, 501, 490 nm, respectively)) (Hai et al., 2020; Jacobs and Neitz, 1989). In contrast, the last common ancestor to walrus and sea lions retained the ancestral Y277, giving rise to comparably longer-wavelength LWS sensitivity (Fig. 10) (Crescitelli, 1958; Fasick and Robinson, 2000; Lavigne and Ronald, 1975; Levenson et al., 2006a; Lythgoe and Dartnall, 1970; Southall et al., 2002). In goats, re-substitution of F277Y back towards the ancestral state recovered longer-wavelength sensitive LWS (Fig. 10) (Jacobs et al., 1998; Yokoyama and Radlwimmer, 1999).

In addition to site 277, other sites are also significant determinants of LWS/MWS wavelength sensitivity. In contrast to other placental mammals, most rodents, and the Lagomorpha have relatively short wavelength sensitive LWS opsins (Fig. 10). The last common ancestor of these animals likely gained H197Y and A308S^{OH} substitutions, which downshifted LWS sensitivity (Fig. 10). However, derived substitutions in the last common ancestor to the grey squirrel and guinea pigs of A180S^{OH} and S^{OH}308A partially recovered LWS sensitivity towards longer wavelengths. A subsequent T^{OH}285A substitution in the guinea pig lineage underlies a 14 nm decrease in LWS sensitivity between guinea pigs and grey squirrels (Fig. 10) (Blakeslee et al., 1988; Yokoyama and Radlwimmer, 1998), in addition to a downshift in Degu, Alpine marmot, Golden hamster, Jerboa, Mongolian gerbil, and the Elephant shrew LWS/MWS (Hai et al., 2020). Similar to the T^{OH}285A substitution that occurred in many rodents, a derived T^{OH}285A substitution also

down-shifted wavelength sensitivity in the semi-aquatic sea otter (Fig. 10) (Levenson et al., 2006b), in the last common ancestor of the tammar wallaby, quokka and fat-tailed dunnart (Arrese et al., 2002, 2005; Cowing et al., 2008; Deeb et al., 2003; Strachan et al., 2004), the white-tailed deer (Yokoyama and Bernhard Radlwimmer, 1998), and the microbat *M. velifer* (Wang et al., 2004).

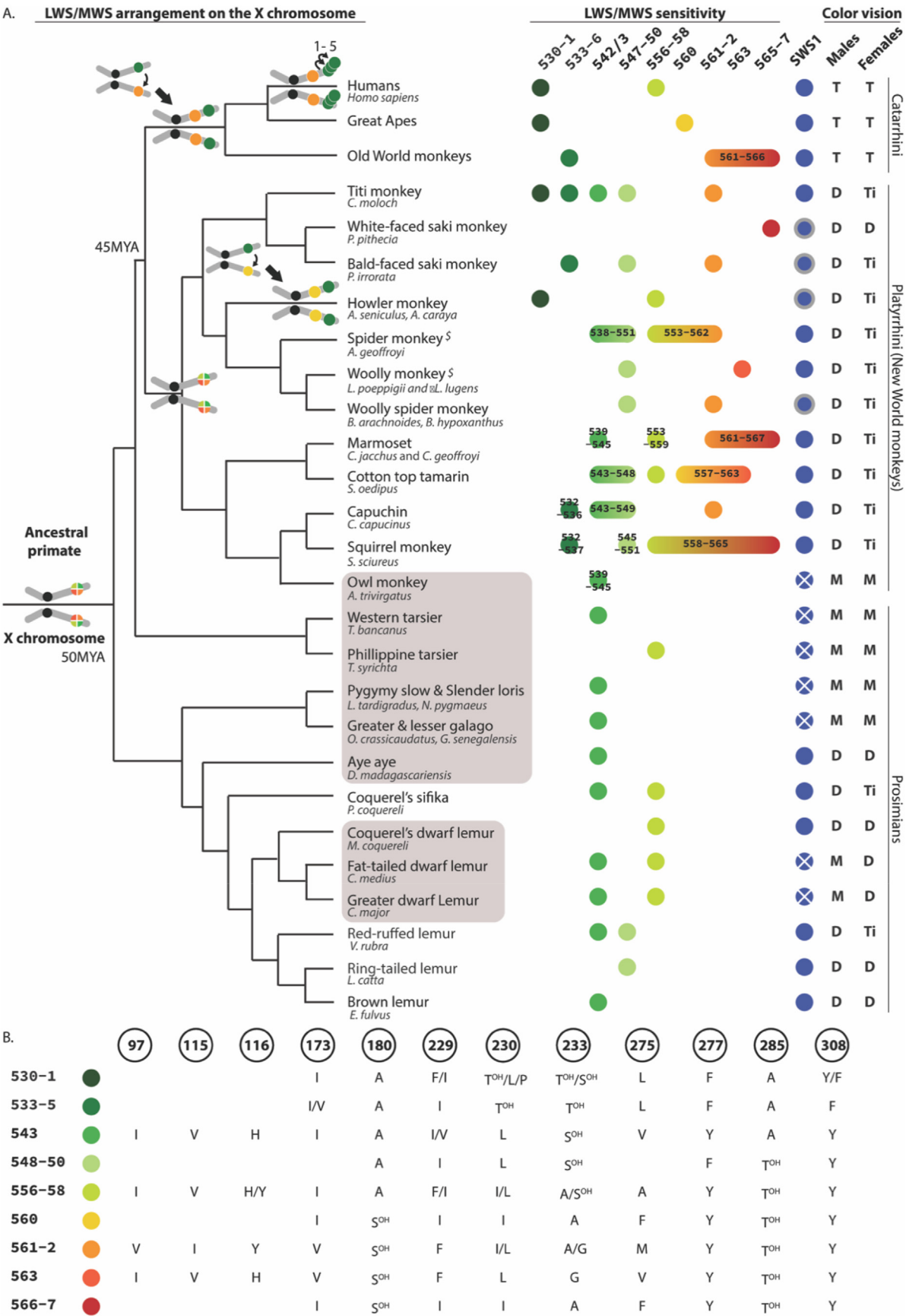
Additional loci outside of the five key sites are required to explain spectral shifts in the LWS/MWS opsins of some species. For example, the unusually down-shifted LWS/MWS opsin of the Mongolian gerbil and the elephant shrew is also attributed to A48T^{OH}, H294Y and S^{OH}107G, S^{OH}134A substitutions, respectively (Hai et al., 2020; Jacobs and Neitz, 1989). However, similar to sites that show clear signs of co-evolution in SWS1 (86 and 90), and Rh1 (292 and 290), natural variation in LWS/MWS at sites 277, 285 and 308 are conserved loci underlying convergence of spectral sensitivity across vertebrates.

In addition to the trends described in non-primate vertebrates above, we find a complex series of changes that altered LWS/MWS sensitivities in primates during evolution. As many as 12 sites likely underlie the multiple LWS/MWS alleles in primate populations (97, 115, 116, 173, 180, 229, 230, 233, 275, 277, 286, 308, Fig. 11B). While the number of amino acid combinations makes their spectral tuning more difficult to untangle, some general trends can be detected. For example, primate LWS/MWS alleles with A180, 277Y, 285A, 308F are sensitive to shorter wavelengths than those with S^{OH}180, 277F, 285T^{OH}, 308Y (Fig. 11B).

What drives changes in spectral tuning of LWS/MWS opsin during evolution? In marsupials, a key driving factor is foraging. The honey possum has a relatively long-wavelength sensitive LWS/555 compared to other marsupials (Fig. 10). This long-wavelength sensitivity provides an advantage in detecting target flowers and discriminating between target and non-target flowers (Sumner et al., 2005). Interestingly, the optimal opsin wavelength sensitivity for these purposes would be at even longer wavelengths (615 nm for detecting target flowers and 640 nm required for discriminating between them). Why then, is the marsupial opsin tuned towards 555 nm light? One argument is that this wavelength is optimal for discerning ripeness of target flowers. Therefore, detecting 555 nm light may provide a “good enough” ability to detect target flowers, whilst enabling high accuracy in determining foods that are the ripest (Sumner et al., 2005).

Similarly, differences in ungulate LWS sensitivities have been linked to feeding behavior. For example, deer are selective browsers who likely benefit from detecting medium light wavelengths. Deer pick out parts of plants for feeding and prioritize nutrient-rich regions such as leaves, fruits, and soft shoots, which differ in coloration from other vegetation. In contrast, bovines are grazers that feed on any accessible grasses, engulfing both indigestible and digestible material at large volumes. This generalized approach to feeding does not require discrimination between different parts of vegetation. Together, the down-shift in LWS sensitivity in deer may have evolved to distinguish between nutrient-rich and nutrient-poor foods.

In addition, spectral tuning of LWS opsins in bats has been linked to non-ocular senses. Some species utilize echolocation, whereas others, such as Old World fruit bats, rely on ocular vision for navigation. The relevance of echolocation on the evolution of vision in bats is under debate. It has been suggested that echolocation pre-dates complex ocular vision and was independently lost in bat some species while vision increased in complexity (Heffner et al., 2001; Pettigrew et al., 1988). A blue-shift in LWS opsin in both echo-locating and non-echolocating bats has been linked to their use of vision to detect ripe fruits, which implies echolocating bats still rely on ocular vision for some functions. Recent analysis identified a link between reliance on echolocation and opsin gene molecular constraint (Gutierrez et al., 2018). For example, less constraint was detected in the LWS opsins of bats that are more reliant on echolocation, and more variation was detected in SWS1 opsins in species that forage among vegetation compared to outside vegetation. Additionally, multiple parallel losses of SWS1 occurred in noctilionoid bats due to pseudogenization (Sadier et al., 2018). The primary driver for



(caption on next page)

Fig. 11. The evolution of LWS/MWS in primates. (A) The evolution of LWS/MWS from the ancestral primate and allelic variation in extant species, as well as presence/absence of functional SWS1 (absence shown as crossed-out grey circle). LWS/MWS gene arrangement on the X chromosome is indicated with schematics in the phylogeny, with the ancestral primate having one LWS/MWS gene. Each circle represents one opsin gene. In the ancestor to Old World monkeys, great apes and humans, an unequal recombination event led to the tandem arrangement of LWS and MWS opsins (shown as two separate LWS/MWS genes in a tandem array) (Dulai et al., 1994, 1999; Vollrath et al., 1988). An independent event in the howler monkey lineage is also thought to have occurred, resulting in a tandem arrangement of LWS-MWS (Silveira et al., 2014). Green through red colored circles indicate LWS/MWS with down-shifted or up-shifted wavelength sensitivities, and do not truly reflect the true spectra the opsin detect. Multicolored circles represent multiple alleles of the LWS/MWS opsin in the population sensitive to different light wavelengths. Text within the opsin schematics indicate peak absorption ranges from different studies (see Supplementary File 1A). Species names in grey boxes indicate those that live nocturnally. Presence or absence of SWS1 is indicated in the right-most column. Violet circles represent known functional SWS1, violet circles with a grey outline represent presumed functional SWS1, and grey crossed out circles represent SWS1 opsins that have been lost/rendered non-functional. (B) The amino acids at key sites in extant primate LWS/MWS opsins. Spectral sensitivities of the opsins for each species, and the amino acids of each opsin were gathered from (Boissinot et al., 1998; Bowmaker et al., 1980; Dulai et al., 1994; Hiramatsu et al., 2005; Ibbotson et al., 1992; Jacobs et al., 1994; Jacobs and Deegan, 2005a, 2003; Kainz et al., 1998; Kawamura et al., 2001; Kawamura and Kubotera, 2003; Merbs and Nathans, 1992; Mollon et al., 1984b; Nathans et al., 1986; Neitz et al., 1991; Shyue et al., 1998; Silveira et al., 2014; Talebi et al., 2006; Tan and Li, 1999; David S. Travis et al., 1988; Williams et al., 1992; Yokoyama, 2000b) (Supplementary File 1A and 1B). Note, opsins with a large range of recorded peak absorptions (± 3 nm) were omitted from this figure for simplicity.

SWS1 in these lineages appears to be diet, specifically target fruit composition. Together, these observations suggest a relationship between echolocation usage and opsin molecular evolution.

Coevolution of LWS and SWS subtypes: In some cases, spectral tuning of LWS opsins is an adaptation to the loss of short-wavelength-sensitive opsin subtypes. For example, the elephant shark *Callorhynchus milii* have duplicated and spectrally distinct LWS1/499 and LWS2/548 (as well as Rh1/496 and Rh2/442 (Davies et al., 2009a)). Analysis of the LWS1 and LWS2 protein sequence indicates that the spectral sensitivity of LWS2 is within the expected range based on the five site rule (Yokoyama and Bernhard Radlwimmer, 1998), while LWS1 sensitivity is significantly down-shifted (Davies et al., 2009a). This relatively short LWS1 peak sensitivity is thought to be caused by an epistatic interaction between H181 and S^{OH}292 (Davies et al., 2009a). In contrast to LWS proteins with positively charged amino acids at site 292 (such as A292), the negatively charged hydroxyl group of the elephant shark LWS1 S^{OH}292 is thought to natively interact with the chloride ion binding site at site H181 to lower the light wavelength required for photoactivation. This scenario has also been detected in LWS opsins of other aquatic vertebrates, such as the pilot whale, bottle nose dolphin, and harbour porpoise (Fasick et al., 1998; Newman and Robinson, 2005). It is possible that the down-shift of LWS1 is an adaptation in response to the loss of functional SWS opsins in this lineage, since living at a depth of 200–500 m, this species is likely predominantly exposed to short-wavelengths of light.

The co-evolution of LWS alongside other opsin subtypes has been described in lizards. In geckos, male coloration indicates dominance to females (Ellingson et al., 1995). Males with more yellow coloration out-compete red males during male-male encounters, even when they are comparable in size, and therefore tend to inhabit optimal territories. Correspondingly, female geckos have a preference for yellow males relative to red males, with yellow males being selected for copulation more frequently (Ellingson, 1994). The optimal "dominance signal" in geckos, and also the best hue discrimination, is at the intersection between the absorbance spectrum of LWS and Rh2 opsins (in the yellow-light wavelengths) (Ellingson, 1994). With these mating systems in place, it is likely that sexual selection pressures constrain the molecular divergence of LWS and Rh2 in tandem.

3.3. The loss, gain and retention of opsin genes

Since the last common jawed vertebrate ancestor, which likely contained Rh1, Rh2, SWS1, SWS2 and LWS, opsin gene losses and duplications gave rise to a diverse array of visual sensitivities in extant species (Fig. 2).

Considering past evolutionary selection pressures: The opsin profiles of extant species are a product of selection pressures experienced in both recent and distant evolutionary history. We must consider the photic environments experienced by ancestral species to understand changes in opsin profiles. Based on the current photic environment alone, opsin

profiles of extant species may appear surprising. One example is observed in mammals, where many diurnal species have limited color vision ability. It has been suggested that early mammals adopted nocturnal lifestyles to avoid predation by archosaur species that were most active during the daytime (Walls, 1942), though evidence for this hypothesis is lacking. Instead, nocturnality appears to be the ancestral mammalian state, predating the birth of the mammalian lineage and occurring in synapsids (Angielczyk and Schmitz, 2014). Nocturnality or diurnality in these synapsids was linked to their diet, with carnivorous species being overwhelmingly nocturnal compared to herbivores. Many extant mammals remain nocturnal, and substantial opsin gene losses occurred during this nocturnal bottleneck, resulting in both nocturnal and diurnal species remaining dichromatic with SWS1/360–460 and LWS/507–560 (Supplementary File 1A). As Rh2 is absent in most mammalian species, it is likely that this opsin was lost very early in the mammalian lineage (Wakefield et al., 2008).

Opsin gene losses in non-visual vertebrates: Some animals do not primarily depend on the sense of vision for survival. In these animals, it is not surprising that they experienced opsin gene losses. At the base of the vertebrate phylogeny, hagfish provide an example. The anatomies of hagfish changed little since the early stages of vertebrate evolution during the Cambrian era and the end of Triassic era (Hardisty and Potter, 1971b). Hagfish eyes represent a transitional form of the vertebrate-style eye, between the non-image-forming eye of tunicates and the image-forming eye of lampreys (Lamb et al., 2007, 2008). They are bottom-dwelling, cartilaginous fish, whose lateral eyes are hidden beneath a layer of opaque cranial epithelium (Holmberg, 1970, 1971). The hagfish species such as *Eptatretus burgeri* lost multiple cone opsin genes and have retina dominated by Rh expression (Fernholm and Holmberg, 1975). Therefore, in species that depend little on the sense of vision, the visual apparatus is somewhat rudimentary in terms of image formation. Instead, the hagfish eye is generally believed to function primarily as a circadian organ, whose ganglion cells predominantly project to the hypothalamus rather than image-forming regions of the brain, similar to the iPRGCs of the mammalian retina that regulate circadian rhythm (Fritzsch and Collin, 1990).

The effect of low-light environments: In many lineages, convergent loss of opsins is observed in species that live nocturnal lifestyles or live in aquatic environments. Aquatic environments provide dynamic changes in light availability, e.g. deeper water (where shorter wavelengths penetrate), shallow marine water (abundant in light), versus brackish waters (enriched for longer wavelength light). Light in aquatic environments are also affected by algal blooms, increased daylight hours depending on season, and levels of rainfall.

In many aquatic animals, short wavelength-sensitive opsins are commonly lost. Complete loss of SWS1 occurred in deep-sea dwelling cartilaginous fish, such as sharks and the holocephali (SWS1 and SWS2), as well as in aquatic mammals such as cetacea (including bottlenose dolphin, pilot whales, harbor porpoises) and pinnipeds (including walrus, sea lions and seals) (Fasick et al., 1998; Hart, 2020; Levenson et al.,

2006b; Levenson and Dizon, 2003; Newman and Robinson, 2005; Peichl and Moutairou, 1998). These species independently lost SWS1, coupled with the loss of SWS2 in the last common therian ancestor. In dolphins, the loss of functional SWS1 was caused by numerous deletions and a frameshift at position 31 (Fasick et al., 1998; Newman and Robinson, 2005). These SWS1 losses may have resulted from relaxed selection due to dependence on other non-visual senses like echolocation to navigate their environments. Alternatively, some animals may have lost their SWS1 as a selective advantage to improve spatial acuity in aquatic environments to the detriment of color vision. The optimal photopigment configuration for spatial acuity involves patterns of neighboring cone cells with the same spectral sensitivity (Calkins and Sterling, 1999).

Similar to trends detected in aquatic animals, the loss of short-wavelength sensitive opsins is often linked to nocturnality (as reviewed in (Jacobs, 2013)). SWS1 was lost in the last common ancestor to monotremes (platypuses (*Ornithorhynchus anatinus*) and echidnas (*Tachyglossus aculeatus*)), (Fig. 8) (Wakefield et al., 2008). Unlike therian mammals, monotremes retained SWS2, which is more paralogous to the SWS1 opsins of reptiles, birds, and fish, rather than the SWS1 opsins of other mammals. In placental nocturnal mammals, many rodents lost SWS1. For example, the Siberian flying squirrel (*Pteromys volans*) and northern flying squirrel (*Glaucomys sabrinus*) contain a non-functional SWS1 pseudogene (Carvalho et al., 2006). Their retinas are dominated by rods and LWS/512-expressing cones. Similarly, nocturnal procyonids such as raccoons and kinkajous are monochromatic, due to the loss of SWS1 opsin (containing only an LWS/560 and LWS/550–560 respectively). Moreover, in nocturnal primates, the owl monkey (New world monkey, *Aotus trivirgatus*) and the last common ancestor to the prosimian lorises/galagos, fat-tailed dwarf lemurs/greater dwarf lemurs, and tarsier lineages also lost SWS1 opsin (grey shading indicating nocturnal lifestyle, Fig. 8) (Jacobs et al., 1996; Kawamura and Kubotera, 2003, 2004; Talebi et al., 2006; Tan et al., 2005; Tan and Li, 1999; Veilleux and Bolnick, 2009). Similarly, Scolecophidian snakes (including blind snakes and worm snakes) depend little on vision and have burrowing lifestyles where they live in low photic conditions and feed on small insects, mainly termites and ants. Likely caused by the limited need for color vision in low-light conditions, Scolecophidian snakes lost all cone opsin genes and retained only Rh1 (Simões et al., 2015). The relaxed selection on vision caused by limited light conditions also manifests itself in other parts of the ocular machinery. These animals typically contain very small eyes that are covered in scales. As a result, both males and females of these species are monochromatic. These independent losses of SWS1 indicate that SWS1 loss is selected for under low-light conditions.

In some cases, short-wavelength sensitive opsins are retained but their expression level is regulated by the photic environment (Luehrmann et al., 2018). SWS1 and SWS2 change expression based on light exposure in nocturnal and diurnal fishes. A decrease in expression of short-wavelength sensitive opsins could be initiated within a matter of weeks and is likely a mechanism that enables androgenous species to adapt to changing photic environments during migration.

This being said, some aquatic and nocturnal animals retain high acuity color vision. In turtle species such as *Pelodiscus sinensis* that inhabit brackish waters enriched for longer light wavelengths, SWS1 is lost while SWS2 is retained. Many turtles remain trichromatic, expressing Rh1/520, SWS2/440–460, Rh2/540–550 and LWS/620–630 (Baylor and Hodgkin, 1973; Emerling, 2017; Liebman and Granda, 1971; Ohtsuka, 1978, 1985a, 1985b) (Supplementary File 1A). Moreover, some sea turtle species, such as *Chelonia mydas* and *Chrysemys picta*, are tetrachromatic, with rhodopsin and four cone opsins (LWS, SWS1, SWS2, and Rh2) (Emerling, 2017). *C. mydas* are sea turtles that inhabit coastlines, and their hatchlings migrate across the beach towards the ocean upon hatching. *C. mydas* hatchlings have a preference for blue coloration (Hall et al., 2018), which may be linked to their need to migrate towards the ocean upon hatching or the detection of food within the blue background of their environments. Therefore, depending on the behavioral ecology of their nesting, natural selection pressures may have maintained functional

short-wavelength sensitive genes in some aquatic species.

Similarly, some nocturnal animals retained multiple opsins, likely due to additional natural selection pressures. For example, henophidian snakes, including pythons and boas, are non-venomous species that have a burrowing lifestyle and are generally nocturnal (Conant and Collins, 1998). While their photoreceptor compositions reflect this dimly-lit habitat (~90% rods and ~10% cones), these snakes still express multiple cone opsins including SWS1 (Sillman et al., 1999, 2001). *Xenopeltis unicolor*, *Python regius*, and *Boa constrictor imperator* express UV-sensitive SWS1/357–361, LWS/549–551, and Rh1/494–497 (Supplementary File 1A) (Davies et al., 2009b; Sillman et al., 1999, 2001). The retention of color vision in these snakes is likely driven by predatory selection pressures, as these snakes have ambush predatory behaviors where improved contrast detection may be beneficial. In contrast, the previously mentioned Scolecophidian snakes similarly display nocturnal lifestyles, but have lost rods and do not exhibit ambush predatory behaviors.

The effect of diurnality: Conversely, high levels of photic exposure in the environment drove the loss of rod photoreceptors and functional Rh1 genes in some species. For example, in diurnal sauropsids, such as some snakes (Walls, 1942) and many lizards (Barbour et al., 2002; Crescitelli et al., 1977; Kawamura and Yokoyama, 1996; Provencio et al., 1992; Röhl, 2001; Walls, 1942), the retina is cone-dominant. For example, the diurnal gecko (*Gonatodes albogularis*) and dragon lizard (*Ctenophorus ornatus*) completely lack rod photoreceptors and only have single cones and double cones within their retina (Barbour et al., 2002; Crescitelli et al., 1977; Kawamura and Yokoyama, 1996; Provencio et al., 1992; Röhl, 2001; Walls, 1942). In these cone cells, geckos and dragon lizards express cone opsin genes (*G. albogularis* express SWS1/362, LWS/542 and Rh2/475 (Ellingson et al., 1995; Kojima et al., 1992; Loew, 1994) and dragon lizards express SWS2/440, MWS/493, and LWS/571 (Barbour et al., 2002) (Supplementary File 1A)). Similarly, caenophidians *Thamnophis sirtalis* and *Thamnophis marciatus* snakes have only cone subtypes, including large single and double cones that express LWS/554, small single cones that express SWS1/360, and small single cones that express Rh1/482 (Jacobs et al., 1992; Sillman et al., 1997; Wong, 1989) (Supplementary File 1A). In some species, such as *Thamnophis proximus*, Rh1 is blue shifted, towards the standard absorption spectrum of the Rh2 opsin, which has been generally lost in snakes (Bhattacharyya et al., 2017; Hauzman et al., 2014, 2017; Katti et al., 2018; Schott et al., 2016; Simões et al., 2015, 2016). The loss of rod photoreceptors in these species, as well as the divergence of Rh1 rendering it either non-functional or more cone-opsin-like, is likely convergent evolution caused by their diurnal lifestyles.

Balancing selection of opsin subtypes: In contrast to the opsin losses described above, multiple opsin subtypes appear to be maintained due to balancing selection in some cases. The taxonomic order of Primates contains an interesting example. The primate lineage includes Old World monkeys (e.g. macaques and baboons), New World monkeys (e.g. platyrrhines such as marmosets, howler monkeys, spider monkeys, capuchins), prosimians (e.g. lemurs, lorises, and tarsiers), and hominoid lineages (e.g. humans and great apes such as chimpanzees and gorillas). In contrast to other placental mammals, some primate species have the potential to be trichromatic (Bowmaker et al., 1991; Dulai et al., 1999; Ibbotson et al., 1992; Nathans et al., 1986). The prevalence of trichromacy varies both within and between primate species and has evolved through different genetic mechanisms.

The current hypothesis suggests that the last common ancestor of extant primates had the potential for trichromacy. This animal had only one X-linked MWS/LWS gene, as well as one autosomal SWS1 gene (Fig. 11A). It is thought that trichromacy was achieved through the divergence of MWS/LWS alleles, giving rise to multiple, spectrally distinct alleles within the ancestral gene pool (Fig. 11A). Females heterozygous for the X-linked MWS/LWS gene were capable of trichromatic vision, since random X-inactivation within photoreceptors across the retina led to expression of different MWS/LWS alleles, in addition to the SWS1 gene. This mechanism of achieving trichromacy through X-

inactivation appears to be conserved in many New World monkeys and some prosimians (see "Ti", Fig. 11A, Supplementary File 1A) (Boissinot et al., 1998; Caine and Mundy, 2000; Hiramatsu et al., 2008, 2005, 2004; Jacobs et al., 2002; Jacobs et al., 2001; Jacobs and Deegan, 2005b, 2003; Kainz et al., 1998; Kawamura et al., 2001; Kawamura and Kubotera, 2003; Mollon et al., 1984a; Travis et al., 1988; Veilleux and Bolnick, 2009; Williams et al., 1992).

A direct consequence of using spectrally diverse, X-linked LWS/MWS alleles to generate trichromacy is that any two individuals of the same species could have very different color vision acuities. Females that by chance inherit LWS/MWS alleles with relatively similar spectral sensitivities would presumably have diminished color discrimination capabilities compared to those with LWS/MWS alleles at opposite ends of the red-green spectrum. Moreover, due to the X-linked nature of LWS/MWS alleles, monochromacy has arisen in males of prosimian species (Fig. 11A). For example, the fat-tailed dwarf lemur and the greater dwarf lemur both lack SWS1 opsin, but retained two LWS/MWS alleles within their populations (Tan et al., 2005; Tan and Li, 1999). Since males only inherit one X chromosome, males of these species are monochromatic, while females have the potential for dichromacy through X-inactivation. Therefore, color perception is very different between male and females of these species, which presumably has consequences for social interactions.

How the multiple LWS/MWS alleles are maintained in these primate populations is an interesting question. The selection pressures that maintain the multiple LWS/MWS alleles have been linked to a heterozygote advantage, with trichromatic females outperforming dichromatic individuals in certain foraging tasks (Abreu et al., 2019; Caine and Mundy, 2000; Melin et al., 2017; Smith et al., 2012). For example, in the common marmoset (*Callithrix jacchus*), trichromats consistently outperform dichromats in their rate of insect capture (Abreu et al., 2019), suggesting that intraspecific variation in opsin subtypes persists because of a fitness benefit for heterozygotes. Therefore, alleles may be maintained at equal frequencies to increase the probability of female heterozygotes (Jacobs and Neitz, 1987). However, dichromatic marmosets outperformed trichromats when identifying cryptic targets, such as hidden, camouflaged predators or food sources (Caine et al., 2010; Melin et al., 2017; Smith et al., 2012), suggesting that balancing selection may maintain the polymorphism. Alternatively, intraspecific variation may be maintained due to mutual benefits of mixed tri- and dichromatic groups. Mixed populations of Verreux's sifaka (*Propithecus verreuxi*) have higher body masses than groups containing only dichromats (Veilleux et al., 2016). In contrast to these examples of heterozygote advantage, the foraging rate in *C. capucinus* displayed no significant long-term fitness benefits (i.e. fertility, offspring survival, maternal survival) for trichromatic females (Fedigan et al., 2014), suggesting that having different LWS/MWS alleles may be beneficial for some species in some conditions but not for others.

Opsin duplication and recombination events: Duplication events can facilitate the gain of biological function and adaptive diversification. Both single-gene and whole-gene duplications have contributed to the diversification of opsin genes throughout vertebrates (Fig. 2). In general, it is thought that a common mechanism involves subfunctionalization (Force et al., 1999) or neofunctionalization. If a duplicated gene had no sequence divergence and was expressed in the same spatial and temporal manner, mutations would likely accumulate in a redundant copy and render the gene non-functional. However, often, duplicated opsins accumulate changes that enable them to detect new light wavelengths that the original copy was not sensitive to, or be expressed differently throughout an animal's lifetime. This may involve novel functions for the opsin gene, or subfunctionalization, whereby the role of the original gene is now subdivided between the duplicated copies.

Primates provide an interesting example of neofunctionalisation of opsin genes. Unlike most New World monkeys (described above), the howler monkeys, *Alouatta seniculus*, and *Alouatta caraya* have distinct MWS and LWS genes, which are thought to have arisen by a

recombination event between LWS/MWS X-linked alleles, coupled with accumulated variation that tuned them towards different wavelengths. This event created a tandem arrangement of LWS and MWS opsins on the X chromosome, including both opsin coding regions and a regulatory region known as the "Locus Control Region" (LCRs, discussed more below) (Silveira et al., 2014). Thus, the LWS/MWS gene array is arranged as LCR-LWS-LCR-MWS. Similar to howler monkeys, catarrhines (hominoids and Old World monkeys) have LWS and MWS opsin genes that are arranged in tandem on the X chromosome. It is thought that this arrangement is derived from a recombination event that occurred in their last common ancestor around 40 MYA, independently from the event in the howler monkey lineage (Fig. 11A) (Dulai et al., 1994, 1999; Vollrath et al., 1988). In contrast to howler monkeys, the recombination event in catarrhines occurred between the LWS and MWS allelic variants, resulting in the generation of a haplotype of LCR-LWS-MWS (Wang et al., 1992, 1999).

In addition, recombination events led to dramatic variation in the copy numbers of MWS and LWS opsin gene in humans (Nathans et al., 1986). MWS opsin number in humans varies between 1 to 5 copies, with a modal number of 2 (Drummond-Borg et al., 1989). However, sequences of these duplicated MWS genes do not differ in spectral tuning sites and are not thought to be highly expressed. This being said, "anomalous trichromacy" is not uncommon in humans (whereby the LWS or MWS opsins are spectrally shifted due to molecular changes). Females heterozygous for a spectrally shifted opsin has been proposed to be tetrachromatic, due to X-inactivation in a manner similar to female New World monkey trichromacy. However, females heterozygous for these shifted LWS or MWS alleles do not exhibit tetrachromacy in behavioral analysis (Jordan et al., 2010). Perhaps the mechanisms responsible for generating color opponent signals of cone photoreceptors are not in place in humans to extract tetrachromatic information.

Generally, catarrhines express both LWS and MWS, as well as SWS1 (SWS1/424–434). These three opsins are exclusively expressed in different cone subtypes, making both males and females "true trichromats". (see "T" in Fig. 11A, Supplementary File 1A). As a result, these species likely experience significantly lower levels of sexual dimorphism in color vision than New World monkey and prosimian species.

What selection pressures drove the evolution of trichromacy in this lineage? One theory considers the benefit of all individuals detecting longer-light wavelengths. A bias towards red coloration is thought to predate the evolution of true trichromacy in catarrhines (Fernandez and Morris, 2007). This bias was likely driven through sexual selection, as primate species with gregarious mating systems are more likely to evolve red pelage and reddish skin compared to non-gregarious species. In some Old World monkeys, red coloration has been proposed to signal mate quality to females, giving rise to this preference (Setchell et al., 2006; Waits et al., 2003).

If these selection pressures are key drivers in the evolution of true trichromacy, why has it not arisen in New World primates or prosimians with gregarious mating systems? One interesting hypothesis is that the heavier dependence on olfactory communication in prosimians and some New World monkeys obviated the drive to true trichromacy (Aujard, 1997; Gilad et al., 2004). The vomeronasal organ is an organ that is commonly used in animals for chemical (olfactory) communication. The loss of this organ in catarrhines coincides with the gain of true trichromacy (Gilad et al., 2004). Presumably, this loss occurred due to a switch in the emphasis of olfactory communication to visual communication in these animals. However, not all primates without the vomeronasal organ are true trichromats and this is likely one of many potential mechanisms that drove the evolution of this trait.

Trichromacy in marsupials: In contrast to the opsin gene duplication or X-inactivation mechanisms of generating trichromacy in primates, some marsupial species have trichromatic color vision through retention of ancestral opsin genes. While the Australian tammar wallaby, south American opossum, stripe-faced dunnart, grey short-tailed opossum, and fat-tailed dunnart are dichromatic (having lost Rh2 and SWS2, while still

expressing LWS/530–555 and SWS1/360–420 (Deeb et al., 2003; Hunt et al., 2009; Strachan et al., 2004)), the quokka, honey possum, and quenda are trichromatic, having spectrally distinct SWS1/350–420, LWS/538–557 and MWS/502–509 (Supplementary File 1A) (Arrese et al., 2005). Interestingly, this MWS gene is similar to the Rh2 gene found in more basal vertebrates (Arrese et al., 2002), and is likely derived from this subtype.

Opsin gene duplication in anurans: Recent studies detected duplication of LWS in the sex chromosomes of the African bullfrog (Schott et al., 2022). However, unlike the tandem duplication event of LWS that occurred in catarrhines, the duplicated LWS gene sequence was introduced onto the opposing sex chromosome (for example LWS1 on W chromosome and LWS-2 on Z chromosome). This mechanism improved color vision in a similar manner to the allelic variation detected in New World primates (as discussed above). For example, females (ZW) would have both LWS-1 and LWS-2, while males (ZZ) would have LWS-2 only. Therefore, the degree of sexual dimorphism of vision detected in some New World monkeys may not be exclusive to that lineage.

Opsin gene duplication in fishes: Teleost fish are a species-rich vertebrate group that inhabit a wide range of photic environments, from deep-sea to streams. Reflecting this habitat diversity, many fishes, including prominent lab models such as the medaka *Oryzias latipes*, (Matsumoto et al., 2006), zebrafish (*Danio rerio*) (Chinen et al., 2003), and guppies (Archer et al., 1987; Archer and Lythgoe, 1990; Kawamura et al., 2016), have multiple copies of many opsin genes. While some of these opsin copies arose from gene duplication events, a third round of whole genome duplication occurred in the teleost lineage (3R) (Meyer and Van de Peer, 2005). Nevertheless, many duplicated cone opsins were subsequently lost. In a basal teleost lineage, the Osteoglossomorpha, lost all functional copies of Rh2, and instead retained an additional LWS opsin that is green-tuned (Liu et al., 2019). Similar to primates, loss of Rh2 in an ancestral species was followed by LWS duplication and divergence into green-sensitive MWS. Intriguingly, the events that gave rise to duplicated opsins seem to be related to specific subtypes. For example, in ray-finned fishes, Rh1 duplication events were often caused by retrotransposition (retrotranscription of Rh1 mature mRNA which is inserted into the genome), while duplication of cone opsins like Rh2 more often arose from tandem duplication events (Fujiyabu et al., 2019), (although one of the LWS duplicated genes of guppies and killifish was also generated by retrotransposition (Sandkam et al., 2017; Ward et al., 2008)).

The most extreme example of opsin duplications in fishes is in the Silver spinyfin, *Diretmus argenteus*, which contains an astonishing 38 rhodopsin genes as well as two cone opsin genes within the genome (Musilova et al., 2019). Regardless of the way they were generated, why are so many functional copies of Rh1 maintained within fish genomes? In the case of the silver spinyfin, many of the Rh1 proteins are blue-shifted. This neofunctionalization of Rh1 duplicated genes may enable capturing as much residual daylight as possible.

In addition to Rh duplications, cone-opsins are also often duplicated in fishes. For example, polymorphism analysis of guppies identified at least 15 different LWS haplotypes, capable of encoding 7 different proteins (Endler et al., 2001; Hoffmann et al., 2007). More recent analysis detected four LWS opsins (LWS1/571, LWS2/562, LWS3/519, LWS4/516) within the *Poecilia reticulata* genome (Kawamura et al., 2016). Similarly, in cichlids, up to fourteen different LWS haplotypes have been identified, occurring at a range of frequencies in various species. These haplotypes are thought to have arisen in response to the variability in light environments among African lakes (Terai et al., 2002), as well as through "sensory drive", where the sensitivities of females to various light wavelengths co-evolved with the coloration of males (Boughman, 2002; Carleton et al., 2005a; Maan et al., 2006; Terai et al., 2006). Since variant sites within LWS haplotypes are also observed within cichlid species, the polymorphic LWS alleles were likely generated by strong divergent selection pressures prior to the establishment of each species (Terai et al., 2002, 2006).

Spectrally distinct duplicated opsins can be differentially expressed during distinct life stages or sexes. For example, in both male and female guppies that reach sexual maturity, the proportion of photoreceptors expressing SWS2-2 and Rh2-2 opsins decreases compared to juveniles (Fig. 12)(Laver and Taylor, 2011). Instead, the number of photoreceptors expressing LWS-1 (A180) increases, which likely improves the foraging of the adult by allowing better detection of foods containing carotenoids (which are orange/red in hue) (Laver and Taylor, 2011). In females, the number of photoreceptors expressing the A180 allele is slightly higher than in males, and the number of LWS-3(S^{OH}180)-expressing photoreceptors also increases (Fig. 12). As a result of these expression changes, sexually mature female guppies are sensitive to longer light wavelengths than males (Laver and Taylor, 2011). Therefore, female guppies detect the orange/red colored spots that accumulate on their scales, which are formed from the carotenoids found in their foods (Fig. 12B, Step. 1). As the number/color of the spots is correlated with the amount of consumed food/carotenoids, it is thought that these spots serve as a social signal for foraging success (Fig. 12B, Step. 1 and Step. 2) (Endler, 1983). Female guppies prefer mates with larger orange spots and color saturation (Step. 3) (Grether, 2000; Houde, 1987, 2019; Laver and Taylor, 2011; Rodd et al., 2002). Therefore, duplicated alleles likely evolved under natural selection pressures that improved foraging and sexual selection pressures that promoted attraction of a mate/detection. Fishes provide an excellent example of how tandem gene duplication and whole gene duplication enabled divergence of opsin genes under natural and sexual selection pressures.

Counterintuitive opsin evolution: Despite clear links between natural and sexual selection pressures on opsin gene evolution, some cases evade explanation. For example, the sea snakes *Lapemis curtus* and *Acalyptophis peronii* are active during both day and nighttime. Yet, their retinas are exclusively filled with cone photoreceptors, which express SWS1/428–430, MWS/496, and LWS/555–559, having lost rods and Rh1 expression (Hart et al., 2012; Simões et al., 2016). Similarly, the Botta's pocket gopher, *Thomomys bottae*, lives in a virtually lightless environment underground but has a cone-enriched retina, making up 25% of all photoreceptors (Feldman and Phillips, 1984). These cones express UV-sensitive SWS1/367 and MWS/505, along with rods that express Rh1/495 (Williams et al., 2005) (Supplementary File 1A). Retention of a large proportion of cone photoreceptors in these low-light-dwelling animals is counterintuitive and the evolutionary driver remains unclear. Similarly, some viper species such as the South American vipers and African vipers contain retina with extremely sparse cone photoreceptor subtypes that express SWS1/UV-violet or LWS/553–555 (Bittencourt et al., 2019; Gower et al., 2019; Katti et al., 2018; Simões et al., 2015), despite inhabiting well-illuminated environments. Therefore, while natural selection pressures have bought about the changes we would expect in opsin profiles in many instances, this is not always the case. Perhaps these surprising opsin profiles might be explained if we investigated the evolutionary history of their ancestral species.

3.4. Regulation of opsin gene expression

In addition to the molecular evolution of opsin genes, it is also interesting to consider adaptation of the regulatory elements that control their expression. The spatiotemporal and levels of opsin expression are divergent amongst species, allowing adaptation between environments. For example, in New World warbler birds, LWS is expressed at higher levels in birds that inhabit darker, deciduous, and coniferous forests compared to those in more open habitats (Bloch, 2015). Presumably, a higher level of LWS opsin expression in dimly lit environments better captures the limited amount of available light.

How are differences in opsin regulation controlled? In this section, we discuss studies of opsin regulatory regions and external signaling factors that regulate the expression of subtypes in specific temporal and spatial patterns. A key element regulating many opsin subtypes is known as the Locus Control Region (LCR), which highly conserved in sequence

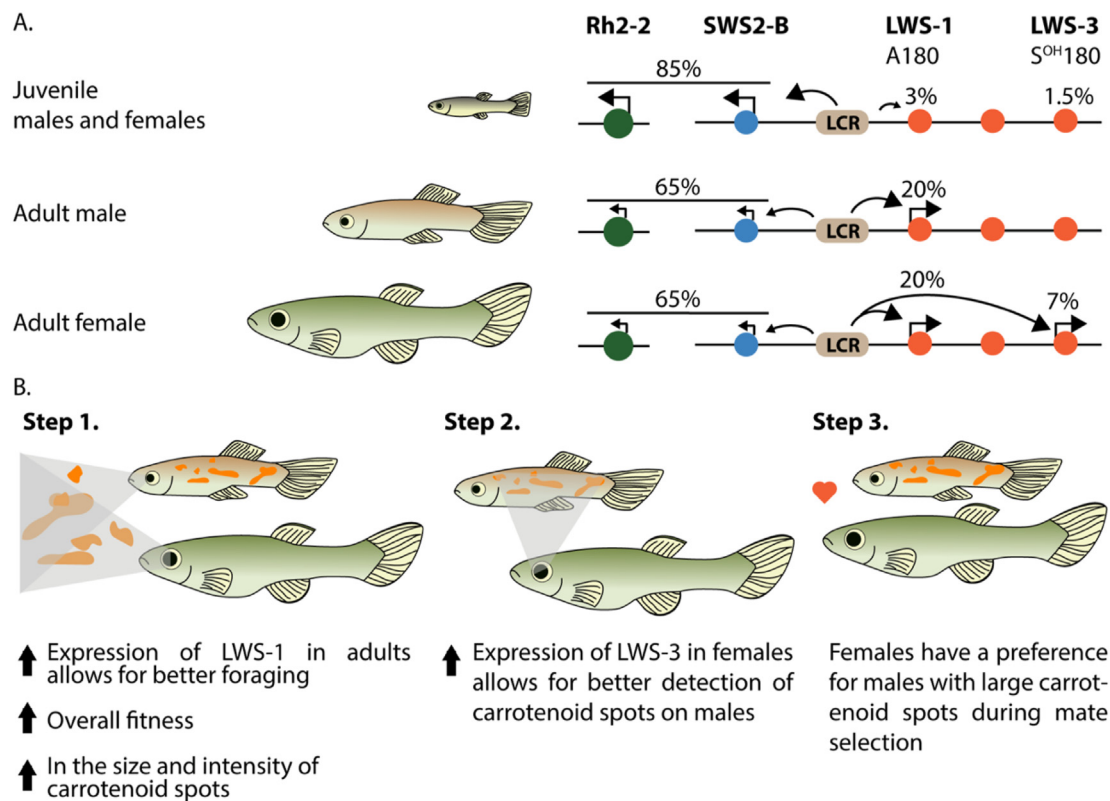


Fig. 12. The natural and sexual selection of SWS2 and LWS opsin expression in guppies. (A). In juvenile guppies, the majority of photoreceptors express Rh2-2 and SWS2 (85%). LWS-1 and LWS-3 are expressed in only 3% and 1.5% of photoreceptors, respectively. In adult males and females, Rh2-2 and SWS2-expressing photoreceptors decrease to 65%, while LWS-1 expressing photoreceptors increase to 20%. Moreover, in females specifically, LWS-3 is upregulated to 7% of photoreceptors. (B) Step 1. With a higher number of photoreceptors expressing LWS-1 opsin, both male and female adult guppies are better able to detect carotenoid-rich food during foraging. The amount of food consumed is positively correlated with the size and intensity of colored spots on the adult scales. Step 2. The upregulation of LWS-3 in females improves their ability to visualize the colored spots on the males. Step 3. Females prefer male mates with larger/more colored spots, using them as a signal for fitness. References used for this figure include (Grether, 2000; Houde, 1987, 2019; Laver and Taylor, 2011; Rodd et al., 2002).

amongst vertebrates. This regulatory element contains multiple binding sites for homeobox genes underlying photoreceptor cell specification, such as *Crx*, in many species (Tsujimura et al., 2007; Wakefield et al., 2008).

Regulation of tandemly duplicated opsins: As mentioned, a key driver of visual function diversification is the duplication of opsin genes. Nevertheless, without the machinery capable of regulating the appropriate temporal and spatial expression of duplicated opsin genes, duplicated opsin sequences may be "unusable" and effectively evolutionarily neutral. How duplicated opsin genes are expressed differs among situations, mainly due to the inherent differences of the duplication event itself. For example, some tandem duplications might involve the gene without the required regulatory elements (such as in the LCR-LWS-MWS array of catarrhines, or the LCR-Rh2 of fishes). In these situations, duplicated genes often compete for the regulatory region.

In the zebrafish LCR-Rh2 gene array, Rh2/467 and Rh2/476 are expressed in the central-dorsal retina, circumscribed by Rh2/488 expression, surrounded by Rh2/505 extending to the ventral retina (Tsujimura et al., 2007). The LCR upstream of the Rh2 opsins drives the expression of all these Rh2 genes, and the level of Rh2 expression appears to be linked to their distance from the LCR sequence, with those located closest to the LCR being expressed more highly (Tsujimura et al., 2007). A secondary LCR known as the LAR ("LWS-activating region") is located between tandemly arranged SWS2 and LWS opsins (Tsujimura et al., 2007, 2010). In this array, the promoters of both LWS opsin genes compete with the upstream LCR (Tsujimura et al., 2010), and differences in their competitiveness may underlie the temporal and spatial differences of LWS duplicated gene expression throughout the retina.

Similarly, following the independent unequal recombination event in the catarrhine lineage, an upstream LCR regulates the expression of

tandemly arranged LWS and MWS genes (Nathans et al., 1986; Smallwood et al., 2002; Wang et al., 1992). The "stochastic model" of LWS or MWS expression proposes that the LCR element randomly loops to either the MWS or LWS promoter to exclusively drive the expression of the opsin within each photoreceptor cell (Nathans et al., 1986; Smallwood et al., 2002; Wang et al., 1999). Experiments using human opsin transgenes in mice showed that the order of opsin genes and their distance from the LCR regulates the proportion of cones expressing each opsin (Hayashi et al., 1999; Yamaguchi et al., 1997). Similar to the duplicated genes in zebrafish, human LWS/MWS promoters seem to have different affinities for regulation by the upstream LCR. The first MWS gene in the array appears to have a greater affinity for activation by the LCR than the LWS promoter (Smallwood et al., 2002). However, distance seems to also play a role, as the LCR appears incapable of driving expression of the downstream duplicated MWS genes. Despite repeated duplication of MWS, it is likely to be evolutionarily neutral in terms of vision. This contrasts with the LCR in fishes, which is capable of regulating the expression of many duplicated opsin genes.

In some cases, such as in the New World howler monkey, opsin genes have undergone tandem duplication alongside their regulatory elements. The unequal recombination event in a female heterozygous for LWS and MWS alleles that generated a haplotype of LCR-LWS-LCR-MWS in the howler monkey is one example (Silveira et al., 2014). Each LCR presumably drives the expression of the downstream LWS/MWS opsin gene, although this has never been experimentally tested.

Bidirectional enhancers: Due to the location of a regulatory element between the SWS2 and LWS gene array in many animals, it was proposed that this LCR ancestrally acted as a bidirectional enhancer (Wakefield et al., 2008). Following the loss of SWS2 opsin in the last common

ancestor to therians, most therian LCR regulate the expression of a single LWS/MWS opsin. In fishes, which retained SWS2, the LCR was thought to still act bidirectionally to regulate the expression of both SWS2 and LWS gene arrays. However, it has been shown that SWS2 expression is regulated by an independent regulatory region (Takechi et al., 2008). A small region 5' of SWS2 contains four OTX binding sequences that are both necessary and sufficient to drive SWS2 expression within the correct cone subtype (Takechi et al., 2008). Moreover, this region represses SWS2 expression within a different cone subtype that normally expresses SWS1.

Retinoic acid and thyroid hormone signaling regulate opsin expression: In addition to genomic distance, exogenous factors also regulate the choice between expression of tandemly arranged opsin genes. Within the zebrafish LAR-LWS opsin gene array (LAR = "Locus Activating Region", a type of LCR), retinoic acid (RA) upregulates the first LWS gene and downregulates the distal LWS gene (Mitchell et al., 2015). This is facilitated by the presence of Retinoic Acid Regulatory Elements ("RAREs") within the LCR sequence and the LWS gene loci, which are binding sites required for the gene expression regulation of RA signaling (Mitchell et al., 2015).

Similarly, we studied human retinal organoids and found that RA regulates the expression of tandemly arranged LWS or MWS opsins in humans (Hadyniak et al., 2021). Dynamic levels of RA over time likely explains temporal differences in LWS and MWS expression in humans, with MWS expression preceding LWS expression in fetal human retinal development (Hadyniak et al., 2021). These findings support a "temporal model" of LWS and MWS expression regulation in humans. These studies also explain spatial differences in LWS and MWS expression, as the early-developing central retina contains a mix of MWS- and LWS-expressing cones compared to the periphery that is enriched for LWS-expressing cones (Carroll et al., 2002; Hagstrom et al., 1998; Xiao and Hendrickson, 2000) (Hadyniak et al., 2021). Differences in LWS- versus MWS-expressing cones in humans were associated with polymorphisms in genes involved in RA signaling (Hadyniak et al., 2021).

In addition to RA, thyroid hormone (TH) signaling regulates the expression of opsins in vertebrates (see (McNerney and Johnston, 2021) for a detailed review). In fish, TH signaling regulates the expression of tandemly arranged opsin genes and promotes the expression of longer-wavelength-sensitive opsins (Mackin et al., 2019; Suzuki et al., 2013). In addition, pioneering work in mice indicated a role of TH signaling in promoting MWS expression over SWS (Applebury et al., 2007; Eldred et al., 2020; Ng et al., 2001). Similarly, in chickens, TH regulate the differentiation of photoreceptors in spatiotemporal waves (Trimarchi et al., 2008), and the promotes the expression of LWS/MWS opsin over SWS (Vancamp et al., 2019). Our studies of human retinal organoids revealed that TH promotes the specification of the longer-wavelength sensitive cones and the loss of SWS-expressing cones (Eldred et al., 2018). Therefore, across vast evolutionary timescales and developmental systems, the role of thyroid hormone in opsin regulation is heavily conserved.

In some cases, the transient expression of opsin proteins has been attributed to dynamics in TH activity throughout lifecycles (such as in the UV-sensitive SWS1 versus blue-sensitive SWS2 cones of the salmonid retina in juveniles versus adults) (Cheng et al., 2009; Greenblatt et al., 1989; Jones et al., 2002). In humans, many color vision defects involving the misspecification of cone cells has been attributed to conditions involving abnormal TH signaling. However, the specification of cone subtypes and opsin expression appears to be plastic, since treatment with TH in patients experiencing hypothyroidism can partially recover color sensitivity (Cakir et al., 2015; Racheva et al., 2020).

Prolactin regulates opsin expression: Opsin expression is dynamic in species that inhabit different photic environments throughout their lives. For example, opsin expression differs in fishes inhabiting saltwater environments compared to those that have adapted to freshwater environments. In three-spined stickleback retinas, expression of these short-

wavelength sensitive opsins decreases upon freshwater adaptation (Pavlova et al., 2022). Already associated with salt-water and fresh-water transitions, prolactin is thought to underlie these expression differences (Pavlova et al., 2022). Prolactin aids in the modulation of salt-water balance during freshwater transitions that occur during spawning. Upon adaptation to freshwater environments, prolactin levels increase in female brains, while prolactin-like protein decreases in both males in females (Pavlova et al., 2022). In the retina, treatment with exogenous prolactin causes a decrease in SWS2 expression level in three-spined sticklebacks. Therefore, hormones like prolactin regulate multiple mechanisms of salinity adaptation, including opsin expression.

In summary, alongside the divergence of opsin protein-coding regions, *cis* and *trans* regulatory elements also adapt to regulate spatial, temporal and expression level dynamics in developing photoreceptors. In some cases, the signaling molecules that bind to these regulatory elements are ancient, but extremely plastic mechanisms for facilitating opsin gene evolution.

4. Concluding remarks

Opsin sequence divergence underlies variability in color vision acuities within and between vertebrate species. Many amino acid changes that spectrally tune opsins are conserved across vertebrate groups, though overwhelmingly, we can detect trends of convergent evolution for each opsin subtype to tune them towards specific wavelengths (such as sites 86/90 in SWS1, 83/292 in Rh1 and 277/285 in LWS/MWS). The effects of substitutions at different sites are often linked to the interactions between the chromophore and the binding pocket of the opsin, which initiate the downstream cascade upon light sensation.

Both natural and sexual selection pressures underlie the evolution of color vision, targeting the opsin protein-coding sequence as well as the regulatory machinery that regulates their expression. In some cases, the opsin gene repertoire cannot be rationally linked to environmental pressures. Perhaps we have not yet identified selection pressures driving these changes in color perception. Alternatively, sufficient evolutionary time may not have passed for those pressures to be reflected in the genome.

The vast diversity of opsin genetic systems, as well as the external factors that regulate their activity, affirms the importance of color vision in vertebrates. With the advent of technologies that allow for targeted genetic manipulation of single nucleotides, additional key spectral tuning sites and more complex non-additive interactions between them may be revealed. Moreover, this genetic power will improve our understanding of how the molecular evolution of genes involved in sensory systems has shaped behavioral interactions both within and between species. In addition, modern approaches such as the generation of retinal organoids may allow for intriguing cases of evolution of opsin genes to be tested experimentally in derived species, such as primates.

Data availability

No data was used for the research described in the article.

Acknowledgements

J.F.D.H. was supported by EMBO ALTF 318-2021. R.J.J. was supported by the National Eye Institute R01EY030872 and the BrightFocus foundation G2019300.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ydbio.2022.10.014>.

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